MEATCHECKER: AN ANDROID-BASED MOBILE APPLICATION FOR PORK MEAT GRADING UTILIZING YOLOv8 ALGORITHM

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The Researchers

DEDICATION

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M.M.D

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ABSTRACT

MeatChecker: An Android-Based Mobile Application for Pork Meat Grading Utilizing YOLOv8 Algorithm was developed, which would help consumers and sellers to recognize pork meat cuts and understand how fresh the cuts are. Poor inspection of pork meat quality, if not done properly, may lead to food safety risks. The goal of this app is to offer a quality assessment system that is reliable and easy to apply for pork meat images and classify them as Fresh or spoiled with machine learning analysis. Background: This article aimed to build a mobile system that can identify pork meat cuts and evaluate the freshness of cuts, implementing the YOLOv8 model on the app for use in an Android-based system for real-time classification. The application was implemented in Roboflow, Google Colab, Android Development Studio, and Kotlin, as well as TensorFlow. The system reads taken photographs, classifies pork meat cuts, and creates bounding boxes to assign freshness scores and confidence ratings. The Technology Acceptance Model (TAM) would be applied to measure perceived usefulness and ease of use by participants in order to measure user perception. Researchers found the usefulness of the application to be strongly satisfactory, based on ISO 25010 software quality standards. This gives Meatchecker: A Mobile Application For Pork Meat Grading Based on Android, an Application using the Yolov8 Algorithm, the ability to assess whether pork meat is fresh or spoiled easily for the user.

Keywords: *Pork Meat Grading, YOLOv8, Image Classification, Android Application, TensorFlow, Roboflow, Kotlin, Meat Freshness Detection, Machine Learning, Technology Acceptance Model (TAM), ISO 25010*

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DEFINITION OF TERMS

Acceptance Testing	The phase of software testing is conducted to verify if the developed system meets the required specifications, business needs, and user expectations before it is officially released for actual use.
Accuracy	Refers to the degree of conformity or correctness of the output or result of a given process or method, usually measured against an accepted or true value.
Artificial Intelligence	The performance metric in machine learning and data analysis that measures the percentage of correctly predicted outputs in relation to the total number of predictions made by a system.
Android	An open-source mobile operating system developed by Google, designed primarily for touchscreen mobile devices such as smartphones and tablets.
Contrast Limited Adaptive Histogram Equalization	An image processing technique that enhances local image contrast by redistributing lightness values while preventing excessive amplification of noise.
Computer Vision	A subfield of Artificial Intelligence that enables machines to process, interpret, and analyze visual data from the real world using digital images and videos.
Confusion Matrix	A table or matrix used in supervised machine learning to evaluate the accuracy of a classification model, showing the number of true positives, false positives, true negatives, and false negatives.
Dataset	A structured collection of data that is organized and labeled for use in computational tasks such as training

	and testing machine learning models.
Deep Learning	A subset of machine learning within the field of Artificial Intelligence that uses multiple-layered neural networks to automatically learn and extract complex features from large amounts of data, enabling high accuracy in tasks such as image recognition, natural language processing, and decision-making.
Feature Extraction	The process of transforming raw input data into a set of features that can be used as input to a machine learning algorithm.
Freshness Classification	The process of evaluating food quality involves categorizing items based on observable characteristics such as appearance, odor, and texture.
Integration Testing	A software testing process that evaluates the interaction and data exchange between different modules or components within a system.
ISO 25010	An international software quality model that provides a standardized framework for assessing software in terms of functionality, usability, performance, security, maintainability, and portability.
Meat Grading	The process of classifying meat products according to established standards, which typically include factors such as freshness, texture, fat content, and overall appearance.
Mobile Application	A software program designed to run on portable devices such as smartphones and tablets, providing users with specialized functions or services.
Scrum	An agile software development framework that organizes work into short cycles called sprints, promoting teamwork, flexibility, and iterative progress.
System Testing	A phase of software testing that evaluates the complete and integrated

	system to ensure that it functions correctly according to specified requirements.
Technology Acceptance Model	A theoretical framework used to explain how users adopt and accept new technology, based on perceived usefulness and perceived ease of use.
TensorFlow Lite	A lightweight version of the TensorFlow machine learning framework designed to run models efficiently on mobile devices and embedded systems.
Unit Testing	A method of software testing that focuses on verifying the smallest testable parts of a program, such as individual functions or modules, in isolation.
You Only Look Once, Version 8 (YOLOv8)	The latest state-of-the-art YOLO model is suitable for object detection, image classification, and instance segmentation tasks.

CHAPTER 1

INTRODUCTION

Project Context

Promoting the quality and safety of pork meat was a major issue in the food-processing sector. Conventional grading procedures, which rely heavily on manual inspection, are often subjective, laborious, and inaccurate. Even more significantly, if the quality of pork meat is confused, then consumers' health is jeopardized as foodborne illness becomes more probable. It also changes the trust people have in meat sellers and regulators. Furthermore, it would lead to financial losses for farmers and corporations, violations of food safety laws, and a breakdown in the supply chain, which would adversely affect the market and consumer confidence. Polkinghorne & Thompson (2010) stated that nowadays, there is a greater importance being added to the quality and safety of meat grading. Pork is widely available in the Philippines for consumption, but the storage quality, grading of pork, and safety problems of pork are still associated issues within the market (Food and Agriculture Organization, 1991). Dependence on butchers and inspectors is liable to produce variable quality reports (Price, 1982), which may be difficult for consumers and lead to health hazards. Moreover, bacterial contamination and meat spoilage also affect the quality of meat (Atanassova et al., 2013).

Pork is not just the most consumed meat in the Philippines, but also the fundamental ingredient of Filipino cuisine and culture. Kasim (shoulder), Liempo (belly), Pork chop (loin), and Tenderloin (lomo) are the most abundant cuts of pork utilized in domestic meals and indigenous foods nationwide.

Tacitly tender, versatile, and flavorful, these cuts are popular ingredients that anyone who doesn't get this lucky enough is unlikely to use them on a plate. Soaring popularity and use make them suitable for regular meat grading practices. Pork had been chosen over beef, chicken, and other poultry as they are the staple food in Filipino diets and are economically crucial. Also, pork is more susceptible to spoilage than other meats when refrigeration is tight in traditional wet markets. This not only makes grading freshness practical but also necessary. By looking at the four types of pork cuts, this study takes into consideration actual consumption patterns and provides real-life, relevant applications.

According to the statistics of Statista (2025), pork is still the most-consumed meat per capita in the country in 2024, exceeding both poultry and beef. Moreover, Rotecna (2023) explains that the pork industry in the Philippines has taken great strides to recover from the worst effects of the African Swine Fever (ASF) outbreak, supporting the sector's significance in the Philippine food supply and economy. The millions of backyard hog raisers and modest sellers who work in the pork sector account for at least a third of the national food surplus. Further, Veterinaria Digital S.A. (2022) reiterated that the swine trade is a major source of economic activity, noting that the product is still also a "big business in terms of jobs and consumption," even after the losses caused by ASF and other problems in the past. Advanced technologies, like computer vision, deep learning (e.g., YOLO), and spectroscopy, are explored for meat quality assessment around the world. Automated meat-grading systems, which have recently been implemented in developed nations, also provide significant efficiency since much of the quality

testing relies on manual testing. Such old practices still prevail in the Philippines, thus the requirement for an easy and tech-enabled grading system to make pork quality and safety a good priority for customers. Country Pork meat grading in the country is also affected by outdated, unreliable, subjective, and manual inspection of pork meat. Without access to updated grading technologies, small vendors can find it impossible to adhere to these standards. The process is also slow and laborious, which impacts the supply chain. Computer vision and deep learning models like YOLOv8 hold great promise; however, they are unable to be put to use or deployed online in real time due to their computational limitations.

Moreover, uncertain freshness measurement creates potential food spoilage and fraud risks. Currently, there is a specific technology gap for the accurate and effective grading of pork. The bad state of the industry comes from the examination process, as it must be carried out by hand, which causes them to lack regularity, their own personal bias, and errors during grading are more common. Additionally, small-scale vendors and butchers have limited access to new grading tools, so maintaining consistent quality and industry standards has been a challenge for these groups.

Computer vision systems and object detection algorithms (i.e., YOLOv8) have been used in food quality assessment, but mobile-based or immediate delivery of pork meat grading solutions have not come together. As an issue, current systems are either too expensive, require more complex equipment than a large number can support, or use them as only used on large-scale process plants. For this gap in technology, food safety, consumer trust, and fair trade practices are all seriously undermined; this creates

serious problems for food producers that people simply cannot change or even eat. With no easy-to-use automatic system to grade meat, vendors fail to grade their products efficiently, and consumers are left with products that are either misclassified at the retail or of lower quality than normal.

Vendors and butcher firms, in general, might not have an advanced grading system, so they have a difficult time keeping quality standards and controls. Increasing consumer demand for safe, high-quality meat-especially among the pork industry, means a better grading system is needed soon that would produce safer quality meat. Computer vision and deep learning methods such as YOLOv8 have shown great potential for automating food quality control. And yet, most current grading systems are set up for large-scale meat processing operations, without low-cost and low-access solutions. This study is important to examine variations and inefficiencies in conventional pork meat grading, based on human-interpreted examination. Such commercially available systems tend to be expensive and out of reach for small vendors, which leads to a technological gap in the industry.

This project introduces an Android Grading system for Android as a low-cost, high-accuracy, and affordable option by implementing a mobile-based grading system targeted at mobile, which is our target user platform. Through computer vision and YOLOv8-Object Detection technology, this method also makes it possible to have an objective and accurate assessment of meat quality, therefore promoting the enhancement of food safety, consumer confidence, and the effectiveness of the process. This innovation would allow small vendors to create a seamless link with up-to-date grade technology, making quality control less expensive and more

standardized. In the local context, particularly in wet markets and small retailers, pork meat is often exposed for sale without environmental controls, so visual inspection (as opposed to a standardized environmental control) has become the primary method of assessing freshness and quality. People often use color, surface moisture, and looks, while vendors use experience, not calibrated tools. These customs are likely to make the grading inconsistent and contribute to the accidental spread of spoiled meat. The development of a mobile-based grading system in the study locates technology in daily market functions that place a premium on affordability, portability, and user-friendliness. The proposed solution meets national food safety objectives as it enables stronger decision-making right at the point of sale. It also promotes the use of technology by small-scale food vendors, transparency in selling meat, and protection of public health. The incorporation of an Android-based pork meat grading application is therefore more than just technical, and rather closes the economic and social gap, cementing sustainable food initiatives as well as creating confidence in the pork supply chain. This background also signals the need to apply AI at the community level, where most pork transactions are transacted daily. Instead of trying to supplant human judgment, the system becomes a decision-support tool and helps the vendor get smarter with the right answers based on data. It can help minimize buyer/seller disputes, facilitate inspections of marketplaces, and offer educational value through raising the awareness of indicators of meat quality. Finally, mobile technology enables scalability and future integration with cloud databases, reporting, and policy monitoring structures. By integrating cutting-edge computer vision methods with real markets, the

research showcases how innovation can be directly employed for the benefit of consumers, end users, and/or the overall food supply chain. This additional context enhances the study's applicability to national development, food security, and inclusive technological advance. It focuses on feasibility, real-world impact, and adoption in the long run.

Project Objectives

This project aims to create an Android-based application that assists consumers and vendors in making informed decisions by using visual indicators for pork meat grading, powered by the YOLOv8 algorithm. This project specifically aims to:

1. To develop a mobile application that can analyze and classify pork meat cuts and their freshness level, categorizing them as Fresh or Spoiled.
2. To implement the YOLOv8 Android mobile application interface and integrate the YOLOv8 model to enable a real-time pork meat scanner and visual freshness classification for consumers and vendors.
3. Test the application through a software review using ISO 25010.

Project Purpose

The purpose of this project is to develop "MeatChecker," a mobile application designed to assist in grading pork meat accurately and efficiently. The app aims to enhance the traditional manual process by providing a faster, more consistent, and user-friendly tool that ensures improved meat quality assessment. This project would benefit the following:

Vendors. Would gain the ability to determine the quality of their products more correctly and quickly. Having this in place also gives them the confidence to show their pork products in the market, as they know the grading is standardized and consistent. Vendors can build trust with customers, and there is a reputation that comes at the top when a valid grading standard is applied to the system. And, with faster grading, sales transactions would proceed faster, easing things up at the point of sale.

Consumers. Would have greater assurance regarding the quality of pork meat available in the market. The application ensures that only properly graded meat reaches them, reducing the risk of purchasing low-quality or misrepresented products. This, in turn, may result in higher consumer satisfaction, improved public health outcomes, and increased confidence in the meat industry.

Future Researchers. Insights gained in this study would also help serve as the basis of future innovations and improvements in the area of food quality monitoring and mobile app development. They can build off a project's observations, add new features, adapt it to other kinds of meat or food products, or incorporate more sophisticated technologies. So this project provides a foundation for further studies in agricultural technology, mobile computing, and food safety, and empowers others to contribute to solving practical problems we face today through innovation.

Scope and Limitation of the Study

This study focused on the development of MeatChecker, an Android-based mobile application that helps users (for example, meat vendors

and consumers) to identify and grade pork meat according to visual quality indicators scanned through the camera of their mobile phone. A trained YOLOv8 model was used for image acquisition, processing, and classification into Fresh and Spoiled categories.

The mobile application was optimized for performance, developed, and tested on an Android 14 (Upside Down Cake) phone with 8 GB RAM and a Helio G100 octa-core processor & rear camera of 50 megapixels. While tested on a single Android device (Android 14), the application worked well with Android 10 (Quince Tart) and is compatible with all higher versions.

The system utilized HSV color space conversion, Contrast-Limited Adaptive Histogram Equalization (CLAHE), white balance correction, mild brightness enhancement, and color jitter adjustments for enhanced image robustness under varying lighting conditions. By applying these techniques, it achieves better contrast and color normalization, which gives more accurate YOLOv8 model detection and classification results.

The development and evaluation of the application are informed by the ISO 25010 software quality model and the Technology Acceptance Model (TAM). The study was limited to pork chop, liempo, kasim, and tenderloin. Very bright or very dark environments still affect the model accuracy, and pork meat cannot be accurately scanned from all angles by the system. How well the app works is also strongly influenced by the camera quality on a phone and the location of the picture taken. Note: The app only assesses the looks of the meat by observing the appearance of the food on it.

CHAPTER 2

THEORETICAL FRAMEWORK OF THE PROJECT

Review Related Literature

This chapter presents a comprehensive review of the related literature and studies, drawn from reliable sources, including online articles, journals, books, and other trustworthy published materials. It would serve as a guide for the researchers in conducting the study. Additionally, it connects and examines several studies that would help in explaining and understanding the significance of the study.

Meat Grading

Pork grading systems differ from country to country and cater to enhancing product performance as well as satisfying consumer demands. An updated grading system in India was established, whereby pig types received yield and quality grades in relation to the current grading system, which aimed to minimize the deficiencies of the original one (Thomas et al., 2018). Previous research on the quality of Korean pork carcass has reported higher grading to relate to the greater fat content, unsaturated fatty acids, and flavor compounds, resulting in high sensory quality (flavor, juiciness, and tenderness) (Van-Ba Hoa et al., 2019). These results emphasize how impactful grading systems are to market needs and consumer satisfaction.

Computer vision systems (CVS) have become an effective substitute for standard methods in evaluating meat quality, providing gentle, fast, and economical solutions (Taheri-Garavand et al., 2019; Modzelewska-Kapituła & Jun, 2022). These technologies can assess multiple quality attributes, such as

color, acidity (pH), moisture levels, and microbial presence, through digital imaging techniques (Rahman et al., 2020). CVS applications include categorizing fresh and processed meats, examining carcass composition, and evaluating the effects of technological processes on meat characteristics (Modzelewska-Kapituła & Jun, 2022). Different methods have been explored for feature removal and classification, including GLCM and KNN (Hanggara et al., 2021), intense learning and OpenCV (Lee et al., 2022), and thresholding with decision tree organization (Adi et al., 2017). Key quality clues assessed include marbling, meat color, and fat color (Adi et al., 2015). Similar techniques have been applied to separate between beef and pork, with one study reporting over 89% correctness using textural analysis and neural networks (Muhammadiy & Fahmi, 2016). The applications are designed for smartphones, allowing real-time meat quality examination. These systems aim to provide objective and efficient alternatives to traditional visual inspection methods, addressing concerns about meat quality and possible fraud in the market (Muhammadiy & Fahmi, 2016; Adi et al., 2015).

Current improvements include the application of object detection algorithms like YOLOv8 NAS for multi-cuisine quality filtering at service points (Kosuru et al., 2023). These systems might be used in place of manual inspection, increasing efficiency and reliability in food quality evaluation (Ambili M & Chandren, 2021). As these technologies continue to evolve, they are predicted to be used widely in an automated production process to provide quality assurance in a competitive food industry and are also likely to be extended to consumer-grade quality review on mobile devices in the coming years (Külcü, 2018). Advanced learning in particular is potentially

viable for on-the-fly application in the food industry, given its feature learning capabilities (Lin et al., 2022). Harmless technologies such as acoustic analysis, machine vision, electronic nose, and spectral imaging, in concert with high-tech machine learning methods, have brought promising results in food quality evaluation (Lin et al., 2022).

Recent research has focused on enhancing real-time detection of pork meat using YOLO-based models. Several studies have proposed advancements to YOLO architectures for pig detection, including the TR-YOLO model, which includes self-attention mechanisms and large layered kernels (Shihua Pu & Zuohua Liu, 2024), and the YOLOv3-SC model, which merges attention modules and distributed pyramid pooling (Wangli Hao et al., 2022). This progress has led to increased detection, accuracy, and effectiveness. Another study introduced an improved YOLOv3 model utilizing transfer learning and attention methods, achieving an average precision of 94.12% (Dan Yin et al., 2021). In addition, a novel approach based on YOLACT was developed for the real-time detection and tracking of pig body parts, achieving 90% validity and a processing speed of 30.5 fps (Fuen Chen et al., 2020). The following advancements provide more effective monitoring of pig health, behavior, and welfare in intelligent breeding systems.

YOLOv8 and CNN

Recent investigations have investigated computer vision and deep learning approaches used for meat quality assessment and classification. Alcayde et al. (2019) implemented a system employing RGB features to determine pork meat age, with proven quality consistent with ISO 25010

specifications. Convolutional Neural Networks (CNNs) are promising approaches for meat classification, like those reported in Bila et al. (2021) with 97.56% accuracy to differentiate beef from pork using a CNN model with dropout regularization. Bacus (2021) trained neural networks to classify the freshness of pork meat, achieving 87.93% accuracy. Though not specific to pork, Dao et al. (2023) used YOLOv8, a CNN-based model, to inspect the quality of agricultural products, achieving 99.5% mean Average Precision for tomato disease detection. This research emphasizes how computer vision and deep learning methodologies, namely CNNs and YOLOv8, have great promise in improving the meat quality evaluation and classification processes.

Recent works have investigated YOLOv8, a CNN model suitable for object detection. The application of YOLOv8 in chicken tracking for lameness detection was also promising, giving high precision and F1-scores when applied in conjunction with optimizer algorithms, like ADAMW (Triyanto et al., 2024). For the multi-class event classification, YOLOv8 outperformed conventional CNN models with a higher mean average precision (Patel et al., 2024). In the agricultural field, YOLOv8 performed well in tomato quality inspection for mAP of 99.5% with great precision and recall (Dao et al., 2023). Furthermore, YOLOv8 has also been successfully used for emotional facial expression detection, with further improvement expected through the addition of data from the available dataset (Alshammari & Alshammari, 2024). The presented results showcase YOLOv8's broad and effective performance among a variety of domains, like in agriculture, human-computer interaction, as well as application to the human condition, so they show that solving

intricate object detection operations with high precision could be achieved using YOLOv8.

Color and Texture in Meat Freshness

The degree of the good quality of pork meat in shaping the identification of its freshness would be an issue of importance for the customers and consumers worldwide. The color and texture are two of the best physical indicators of pork meat quality and spoilage. Color is one of the most fundamental sensory characteristics of a food and influences consumer acceptance in an instant. In general, food perceptions are based on appearance alone; foods are rejected or selected not based on taste or smell but based on color and appearance (Ramanathan et al., 2020). The color red has been linked to arousal signals by reinforcing the perception that the meat is healthy and tasty, Ramanathan explained.

Texture is also one of the main influences on pork quality. New pork tends to be firm and mildly elastic in texture. As the meat deteriorates, the physical traits of this type of food become visible, becoming more maleficent, slimy, sticky, or moisture-laden, which is common for the detection of microbial activity and spoilage (Resconi et al., 2012). Smooth texture changes are also affected by changes in muscle structure, handling, temperature, and storage conditions, making the visibility of changes in texture difficult. Recent studies show that the appearance of color and texture is not only something that can be sensed sensibly, but that technological methods can be leveraged. Machine vision or image processing approaches, for instance, can study the color variation and texture response for the detection of spoilage or also to

classify meat quality in a non-destructive manner (Zhao et al., 2021). Deep learning and computer vision methods, especially CNN-based architectures, have now become popular to automatically detect surface discoloration, marbling, and texture inconsistencies to evaluate meat freshness. These results demonstrate that color and texture are not only human sensory elements, but physical attributes that 21st-century AI systems can also identify to help quality control, grading, and automated freshness monitoring in meat.

Android Application Development Trends

The development of Android apps has seen a big evolution since its inception. The Android mobile application development ecosystem, being Linux-based, gives developers greater freedom and customization in developing their mobile applications (Anirban Sarkar et al., 2019). It works on compatible devices like smartphones, tablets, and smart TVs (Rajesh Keshavrao Deshmukh et al., 2018). The fact that Android is an open-source platform means that it can easily be customized with ease for developers and is faster to implement, and it has an open user base (Anirban Sarkar et al., 2019). However, development does pose a security challenge, since different applications can be exported to the market without strict security conditions. Notwithstanding, Android is one of the most successful mobile operating systems, which provides a stable environment for development of a variety of applications as well as creative solutions and applications for developers (Rajesh Keshavrao Deshmukh et al., 2018; Anirban Sarkar et al., 2019).

TensorFlow Lite and YOLOv8 for mobile applications

YOLO (You Only Look Once) is a robust and real-time object detection algorithm in the field of machine learning, which is fast and accurate (Rajasekhar, 2024). Convolutional neural networks assess the entire image to determine object locations and categories (Gudala Lavanya & S. Pande, 2023). Image recognition, specifically, is approached as a reverse process. YOLOv8 used a single-step pipeline for the detection and is extensively used in applications, such as video security and autonomous driving (Thrupthi C. P. et al., 2024). Besides, YOLOv8 achieved an average processing time of 0.25 s per frame and a frame rate of 4.2 FPS in a relatively recent study confirming that YOLOv8 has a good performance in real-time processing (Bauyrzhan Berlikozha et al., 2024). Also, by predicting the likely objects within each grid cell, the algorithm used grids as inputs. A peak suppression is now used to enhance accuracy by reducing reuse detections (Bauyrzhan Berlikozha et al., 2024). YOLOv8's versatility and consistent performance in different settings have contributed to its widespread acceptance in academic research and industrial processes (Bauyrzhan Berlikozha et al., 2024). For the Food 101 dataset, our model was able to achieve a fast food classification accuracy of 80.20% (Shaon Bandyopadhyay et al., 2024). Also, in food classification, YOLOv8 models have recently been evaluated for application, and YOLOv8 was found to be a very accurate model.

For instance, for mold detection on food surfaces, it could achieve 100% recall and 98.10% accuracy (Md. Fahad Jubayer et al., 2021), as well as an excellent 99.9% accuracy in separating between fresh and rotten fruits and vegetables (M. Sai et al., 2023).

Moreover, a relative study demonstrated that YOLOv5 performed better than YOLOv8 for food classification, achieving higher accuracy (0.851 vs. 0.67), recall scores (0.836 vs. 0.675), and mAP scores (Sudharson S et al., 2023). Recently, TensorFlow Lite and YOLOv8 were used for mobile-based object recognition applications and showed that they can be applicable in a variety of real-life situations. For example, real-time face recognition was done with YOLOv4 and lightweight versions on smartphones. YOLOv4 TFLite achieved an impressive 96.92% accuracy (Ntzelepi et al., 2022). Transfer learning from TensorFlow to YOLO version 9 has also been applied to acute lymphoblastic cancer detection on cellular phones for smartphone applications (Elhalid & Işık, 2024). Moreover, pre-configured TensorFlow models built with TensorFlow-based models have been employed in Android apps for the use of quantization, pruning, and compression for mobile usage for mobile computing (Kashyap et al., 2023). In addition, authors tried to improve object recognition on IoT by applying TensorFlow Lite and YOLO version 4 to solve crucial problems with respect to energy load and computing power limitation in IoT devices (Praneeth et al., 2023). These two are capable of conducting meaningful and powerful object identification on mobile devices. YOLO and TensorFlow Lite are two great tools for us today.

Synthesis and Relevance of the Study

The article collects information from a variety of sources in order to provide a more complete picture of pork meat grading. Prior studies also demonstrated that different ways to assess meat quality have been utilized for assessing variables including color, texture, and fat content. From the research conducted on this topic as well (and from previous studies), it is

understood that correct grading practices help in maintaining the safety of the food, satisfy the consumer, and promote the meat industry by standardizing quality. The study analyzes the works for key points within the context of what they share, such as objective grading standards, technology, and food inspection, and the challenges encountered in traditional grading methods. This overview has facilitated the clear development of the ideas related to the improvement of meat inspection through the improvement in grading techniques. Nevertheless, previous research also emphasizes the need to evaluate the quality and accuracy of meat to meet industry requirements. Different grading systems for pork meat have been set in place that rank the products based on their characteristics to ensure that products of adequate quality are the only ones to be sold. Traditional methods rely primarily on visual assessment, which may be subjective and can suffer from inconsistency. However, different types of measures (scientific measurements and scientific instrumentation) have been suggested to improve the objectivity and accuracy of grading activities. The objective is to develop a far more consistent grading solution that works for producers and consumers, who are in dire need.

Moreover, practical results from meat grading have been studied in the literature, too. Proper control of quality in pork meat production can increase market value and consumer trust. Well-graded meat products have been proven to lower the chance of contaminated products as they meet safety standards. In addition, developing better grading would reduce loss and ensure that only great meat is being passed on to the consumers, which would be one of the many ways to improve the supply chain. The previous

study's insights can be a solid basis for optimization of meat grading to ensure a more standardized process, which can be beneficial in the whole production process.

Literature Matrix

Table 1. Literature Matrix

Author	Year	Title	Findings
Zhu, L., & Spachos, P.	2021	Support Vector Machine and YOLOv8 for a Mobile Food Grading System	The results of this study revealed that SVM was superior to KNN, RF, and NB in banana ripeness classification ability with 98.50% accuracy when trained with feature vector A (H, V, and LBP features). A second-layer classifier was able to detect defects of ripened bananas with an mAP of 0.8239 and a recall of 91.77%, but precision was lower than ground truth labeling, which was not consistent. This study emphasizes the necessity for a standardized ripeness classification technique to achieve increased accuracy and consistency in future work.
Song, S., Guo, Q., Duan, X., Shi, X., & Liu, Z.	2024	Research on Pork Cut and Freshness Determination Method Based on Computer Vision.	It compares different convolutional neural network models of DenseNet, ResNet, MobileNet, VGG, and EfficientNet, and shows that the DenseNet model has the best accuracy (up to 99.28%) compared to the MobileNet model which is slightly less accurate but more efficient and suitable for mobile applications. Data augmentation provided a

			marked improvement in model performance, and the addition of attention mechanisms helped further improve accuracy. A user-friendly graphical interface is proposed for real-time pork cut classification and freshness assessment.
Zhen, Q., Liwei, D., X., Li, Jiacheng, H., Bin L., & Ao X.	2024	YOLOv8 for defect detection and classification of product images	This results in a defect analysis that gives a faster and more accurate result in the market. We performed experiments on industrial product datasets with YOLOv8 and trained it. Results indicate that this enables a real-time detection with a high accuracy and high for, which together with this method would provide a dramatic enhancement in impact and accuracy to the quality inspection of products using the technique.
Antequera et al.,	2020	Non-destructive methods for quality control of fresh meat are used	These nondestructive methods are a fast and accurate alternative to traditional chemical and visual methods for inspection. Yet, more efficient, economical, and robust solutions for external conditions should be found before commercialization in the meat industry.
Aharari & Mehdipour,	2024	Inventory Control in the Food Industry Using an AI-Driven Precision Counting	YOLOv8 has proved to be more proficient than its predecessor in specific food-related tasks inventory management, food damage detection, and poultry farming management. As compared to the predecessor YOLOv7, YOLOv8 processes images 6.3 times faster.

Álvarez-García, W. Y., Mendoza, L., Muñoz-Vílchez, Y., Núñez-Melgar, D. C., & Quilcate, C.	2024	Implementing artificial intelligence to measure meat quality parameters in local market traceability processes	Use of smart sensors and artificial intelligence enhances the accuracy and efficiency of meat quality evaluation. It can dynamically monitor parameters like intramuscular fat, moisture, colour, pH, etc. These technologies drive traceability and ensure food safety and competitiveness in the local market. Implementation of the latter methods in Peru promotes a systematic quality control that results in safer and higher-quality meat.
Anusha, K., Uma, K., Jayasri, K., Kambham, S., & Dandamudi, S. D.	2024	IoT-Based Food Spoilage Detection Using Machine Learning Techniques	The study explores a proposed IoT food spoilage detection system to address these problems here. IoT sensors and machine learning (ML) algorithms are employed to provide real-time analysis of key parameters, such as temperature, humidity, gas, and the availability of food.
Jo, K., Lee, S., Jeong, S., Lee, D., Jeon, H., & Jung, S.	2023	Developments and applications in hyperspectral imaging–based evaluation of fresh meat quality	This is done by utilizing a combination of imaging and spectroscopy to provide external and chemical characterization of the meat based on hyperspectral imaging. A 3D data cube allows for a more accurate, non-destructive quality assessment.

Antequera, T., Caballero, D., Grassi, S., Uttaro, B., & Pérez-Palacios, T.	2020	Evaluation of fresh meat quality by Hyperspectral Imaging (HSI), Nuclear Magnetic Resonance (NMR), and Magnetic Resonance Imaging (MRI): A review.	For the study of meat quality, near-infrared spectroscopy (NIR) and magnetic resonance imaging (MRI) are great potential methods, especially from a laboratory perspective. Nevertheless, their industrial application still needs to go through the iterative development of quicker imaging for data collection, automatic processing of data, and testing the model for accuracy on the production line.

Conceptual Framework

As shown in Figure 1, different images are taken for liempo, kasim, pork chop, and tenderloin under multiple lighting conditions. With the goal of robustness, various lighting environment simulations were implemented. It allows the model to learn how freshness looks in various real-world situations.

During processing, the collected images undergo several enhancement techniques. One of these is HSV color conversion, which separates color information from brightness, making it easier to analyze meat color. Another essential step is CLAHE (Contrast Limited Adaptive Histogram Equalization), which enhances local contrast and improves image quality in low-light or uneven lighting conditions. The images are also resized and labeled accurately into categories such as fresh and spoiled.

Once preprocessed, the images are passed through a deep learning model during the feature extraction phase. In this case, EfficientNet is used to

extract meaningful features such as color tone, texture, and brightness, which are essential indicators of pork freshness.

Fresh (green) and Spoiled (red). At the output stage, the prediction results appear on a mobile application interface. For instance, when a person scans a piece of pork with his or her phone's camera, the app saves and analyzes an image. The result is given like "Fresh: 81%," along with a color marker for the corresponding item to help the user quickly get a feel for how fresh the meat is.

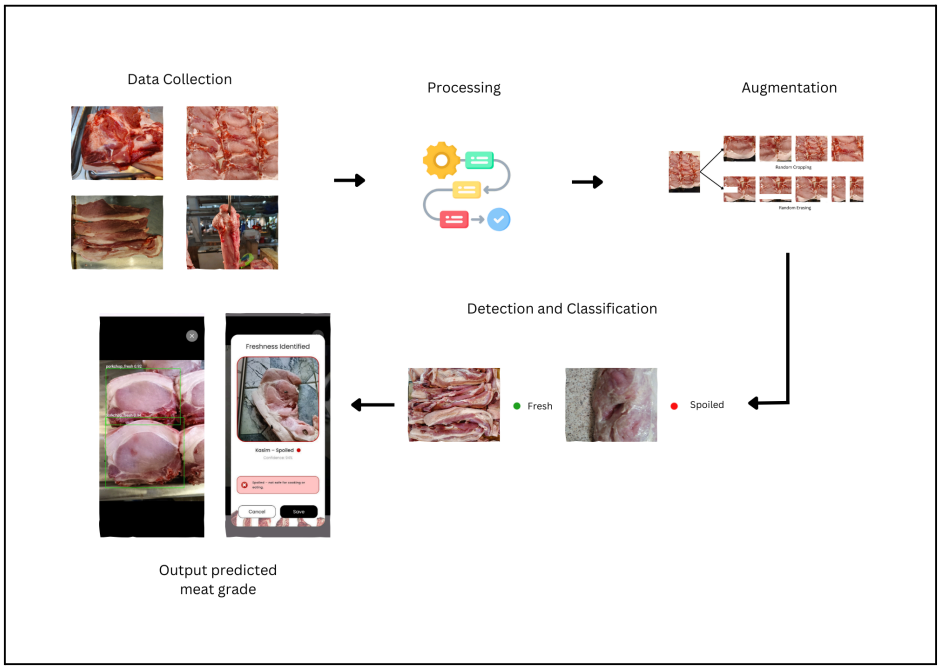


Figure 1. Conceptual Framework

keeps a simple process that facilitates efficient pork meat grading during actual use.

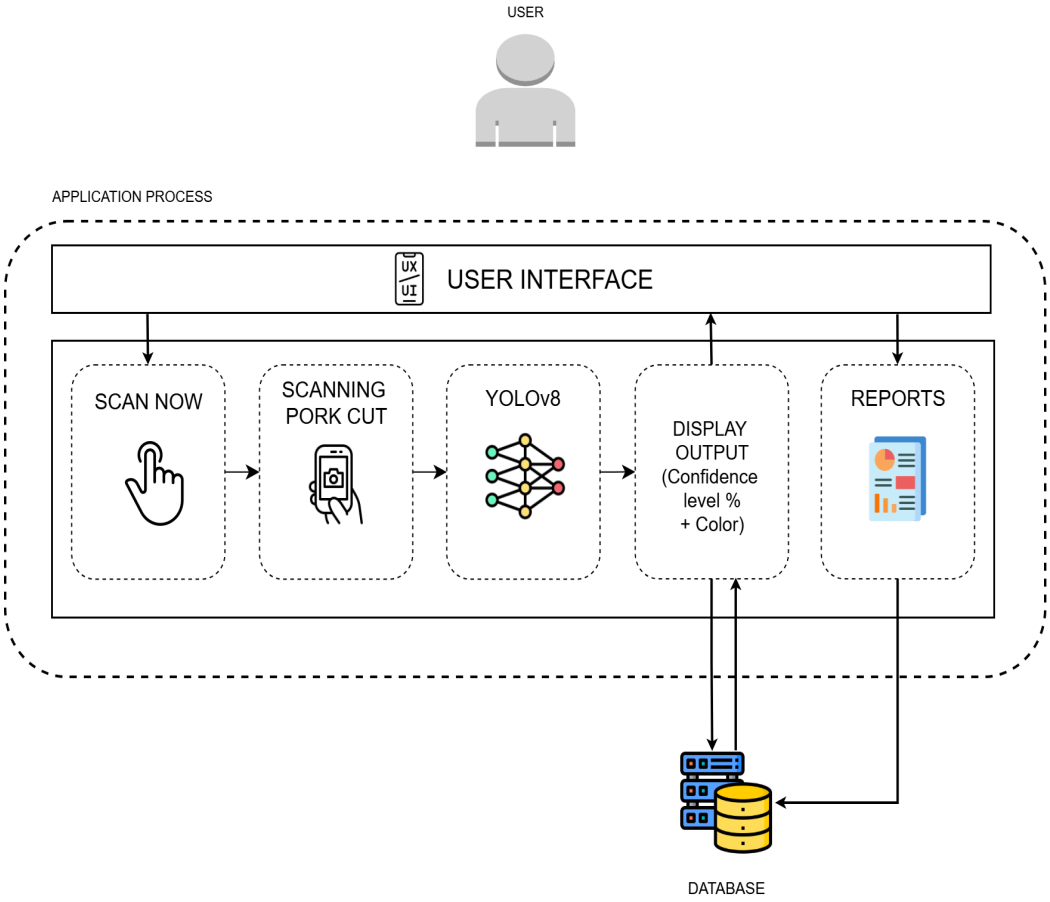


Figure 2. System Architecture

Project Development

The Software Development Life Cycle (SDLC) is a framework that outlines the steps performed throughout each stage of application development. Ensuring that every task is addressed appropriately is crucial. Due to its flexibility and consistency as a medium for adapting to changes and executing the work required to achieve the project's goal, it was used as a project guide.

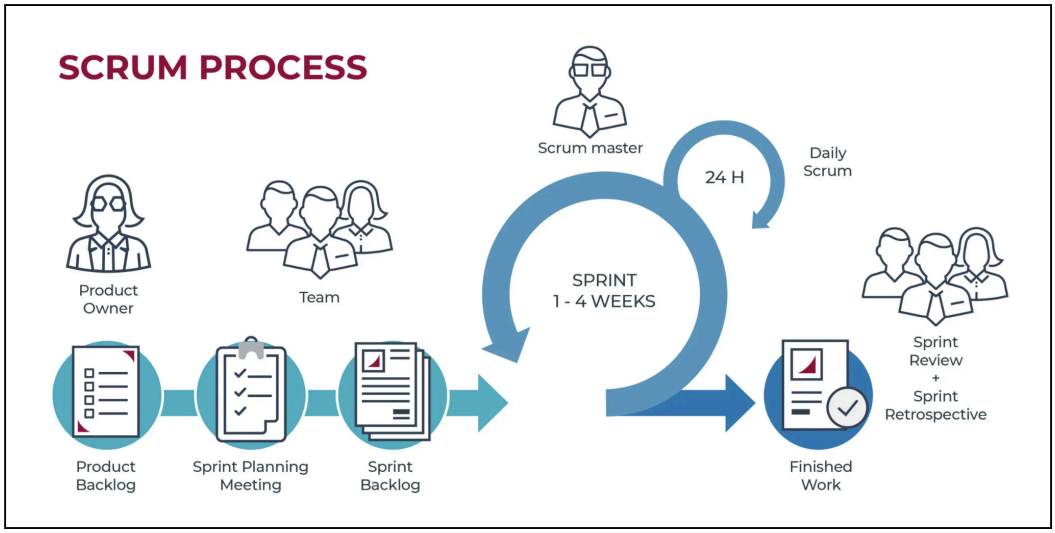


Figure 3. Scrum Methodology

As shown in Figure 3 illustrates the Scrum process begins with the product owner preparing the product backlog, a list of features and requirements for the project. The development team then conducts a sprint planning meeting to select sprint tasks and form the sprint backlog. The sprint typically lasts 1 to 4 weeks, during which the Scrum Master facilitates a daily Scrum held every 24 hours to monitor progress and address concerns. After the sprint is completed, the team produces the finished work, followed by a sprint review and a sprint retrospective to assess results and identify improvements for the next sprint.

Product Backlog

As shown in Table 2, in order to make the MeatChecker app easily usable and more productive, The developers concentrated on the basic user experience. First, the developers understood that it was important to make the app accessible without requiring sign-ins or passwords. They aimed to prevent users from being delayed by log-in and account creation before they were able to scan meat. The onboarding process was streamlined to allow

users to open the app and begin scanning meat immediately. This was a little tweak, but it increased convenience a lot.

Next, the app's biggest function would have to be real-time pork meat freshness scanning through the phone's camera. The developers invested sufficient time in identifying a reliable AI model that utilized YOLOv8 to process images accurately and efficiently. Configuring the AI was one of the more challenging components of the project, requiring the model to be fine-tuned to handle varying angles and lighting conditions for accurate detection of fresh meat. Although the configuration process required additional time, the system was designed to deliver real-time results to users. To help readers easily interpret the results, the developers focused on providing a clear and intuitive visual presentation. Instead of displaying raw data or complex graphs, a simple color-coded system was implemented: green for fresh, yellow for caution, and red for not fresh.

The system was strategically designed so that readers could quickly look up the outputs. It was not overly lengthy to use, yet it significantly improved usability and the overall user experience. An offline mode was also developed to support users when Internet access is unavailable. Since public markets and rural areas often experience unstable connectivity, scanning and result processing were integrated directly into the application to allow it to function offline. This feature required additional development time, particularly for optimizing the app's essential functionalities. As a result, the application operates without continuous online communication, which was crucial in providing a more reliable system. The system introduced a feature that allows

users to store and view their scanning history. This feature was implemented after identifying that users may need to review previous scans, either to keep records of individual meat samples or to compare results over time.

Table 2. Product Backlog

Priority	User	Overall hour/s per day
1	As a user, I want to access the app directly without logging in so that I can use its features easily and without hassle food safety.	1 hour
2	As a user, I want to validate pork meat freshness using my camera in real time so that I can ensure food safety.	2 hrs
3	As a user, I want to receive results with a color-coded freshness indicator so that I can quickly identify meat quality.	1hr
4	As a user, I want an offline mode so that I can scan meat even without internet access.	3hrs
5	As a user, I want to save my scan history so that I can track previous freshness results.	1hr

Sprint Planning

As shown in Table 3, it starts by making sure that visitors are able to open it while using the app without the need to sign in, and therefore the team initialises a welcome screen and creates an easy flow for all guest visitors to easily access the basic app operations. And then they go all in on what's at the heart of the project, that is, how to incorporate YOLOv8 to real-time detect pork meat freshness. That includes training the model, getting it connected to the camera, and making sure the scanning is smooth and fast.

After detection, the next step is to produce results that can be easily recognized by users. The team defines categories for freshness and produces a color-coded interface (green shows fresh, and red shows spoiled) so that people can see what's displayed almost immediately, and they make this necessary decision. For an application that needs internet access but is without trouble, an offline mode is included in this way. This mode requires the necessary files to be stored locally, and it can switch easily between the online and offline modes without any trouble. They try to ensure scan history is saved, so users can go back and check previous results whenever they want. This involves determining the storage of data as well as storing it in a data bank, storing and retrieving past scans, and creating a simple interface to the web app to display the history. This is how the entire sprint moves from just accessible to full integration, enabling the app to be smooth, usable, and useful from the time you open it. Furthermore stresses how the main tasks of every sprint are related to the main components of the MeatChecker application. The key is to remain focused on accessibility, accurate pork freshness detection, and simple output reporting without introducing unwarranted operations. Guest access guarantees instant usability, and YOLOv8 support provides a visual assessment with reliable input from captured pictures. Offline and online modes are managed for scanning continuity only. Scan history storage is restricted to only the information required for scans, image, result, and time, and is done for review and traceability. In summary, the sprint flow exhibits a gradual progression from simple access to stable detection and result management, such that if the application has fulfilled its purpose, it should be effective and consistent.

Table 3. Sprint Backlog

Priority	User	Sprint Activity	Task	Hour/s Day
1	Using an app without signing in	Guest Access Implementation	Design a welcome screen Route guest users to the main features screen	2hrs
2	Validate pork meat freshness using the camera in real-time.	YOLO Integration & Real-Time Detection	Train the YOLOv8 model for meat grading Implement camera-based scanning Optimize real-time processing	7 hrs
3	Display color-coded freshness results for easy identification.	User Interface & Result Interpretation	Define freshness categories Develop UI for displaying the result	8 hrs
4	Offline mode for scanning without internet	Offline Feature Development	Implement local mode storage Optimize offline scanning	5hrs
5	Save scan history for tracking past results.	Data & Storage System	Design scan history storage Display history in the app UI	8 hrs

System Flowchart

As shown in figure 4 is the system flowchart of MeatChecker. The user starts the process with app launch, and it goes to their Home Screen or

Dashboard. In this step, the user presses the Scan Now button from the Home Screen to initiate the detection. The application then turns on the camera, which lets the user take an image of the pork meat. Finally, after the system has captured the image, it enters the Scanning Image step, where the captured image is prepared for analysis. This is then followed by Image Processing to enhance and format the image before sending it to YOLOv8 Detection. The model does object detection and provides bounding box predictions. The output is then subjected to Non-Maximum Suppression (NMS Filtering) to remove the presence of overlaps or low-confidence detections. After the detection is completed, the Display Detection Result screen displays both the detected freshness level and the bounding box information. The user is delivered to the Preview Screen, where they are able to save or cancel the detection. If Retake is selected, the flow proceeds back onto the Camera for yet another scan. If the user selects Save, it is saved to the system's records, so that the Dashboard/Report Screen can access it later. This process ends after the result is successfully stored. The flowchart represents a clear, linear approach, leading the user from detection initiation to result storage in a simplified sequence. All the steps are intended to be user-friendly, straightforward, and intuitive for accurate image-based analysis. This sequence allows a structured approach to image capture, processing, and YOLOv8 detection, which minimizes errors during scanning. Decision points such as retake or save give users control over the final output, while saving results allows consistent tracking via the dashboard. On the whole, this flow displays a controlled and user-centered approach prioritizing reliable pork meat freshness grading.

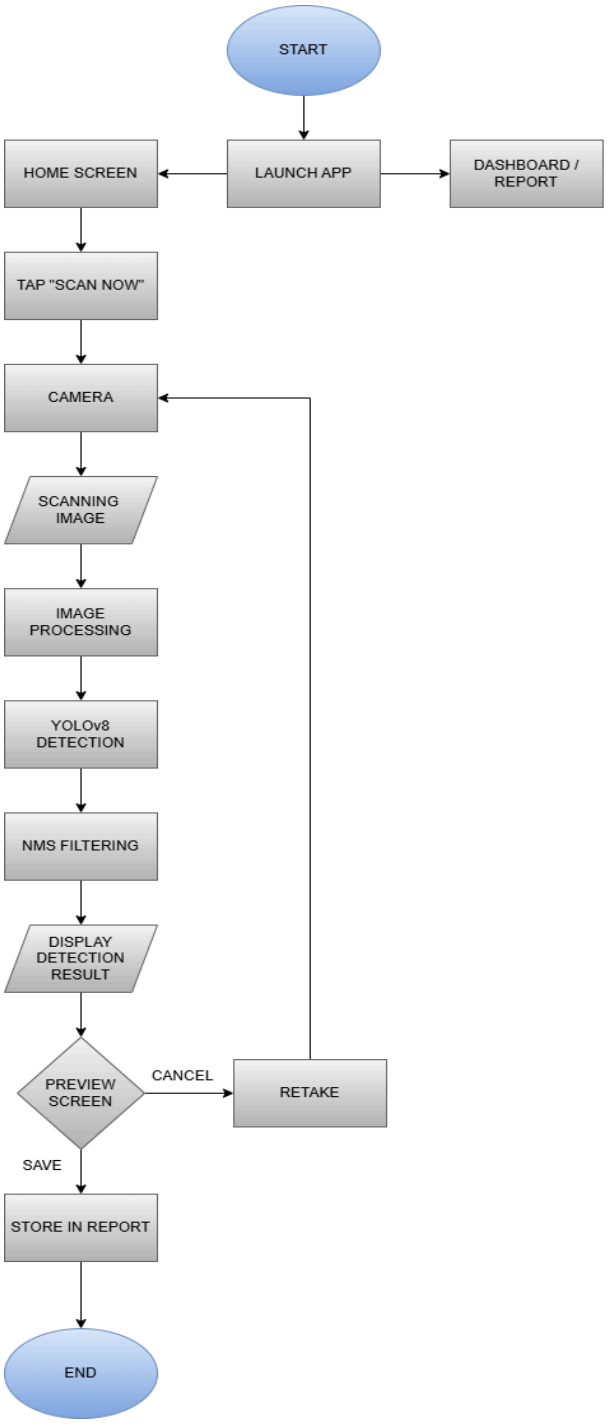


Figure 4. System Flowchart

Materials

This section shows the software and hardware that were used to develop and finish the application.

Software

The software needed for developing the application, along with its features and justification, is listed below.

Roboflow is an online tool that helps you collect, label, organize, and train image datasets for computer vision projects, such as object detection with YOLO.

Google Colab (short for *Google Collaboratory*) is a free online tool that lets you write and run Python code directly in your web browser with no installation needed.

Android Development Studio is best for apps because it works smoothly with the camera, TensorFlow Lite, and YOLOv8 for real-time detection. It also makes building, testing, and designing your app easier and faster.

Kotlin is the official Android language. It's modern, safer, and works perfectly with libraries like CameraX, TensorFlow Lite, and YOLOv8.

TensorFlow is designed to run machine learning models directly on mobile devices, fast and efficiently, even offline. It works perfectly with Android, supports YOLOv8 models, and allows real-time detection without needing an internet connection.

Hardware

As shown in Table 4, the pork meat freshness detection mobile application was developed to work on Android hardware supporting Android 14 or later, capable of working with modern machine learning libraries,

introducing advanced security mechanisms, and optimizing the overall system. The smartphone in this study has 8 GB of RAM, which is good enough to process the images in real time, for YOLO model inference, and for multitasking without causing noticeable delay or system breakdown. The Helio G100 processor provides reliable performance for mobile computer vision tasks, allowing for efficient execution of model operations and rapid response times when performing scanning and classification. The device also uses a high-resolution 50-megapixel rear camera, which is able to provide clear, exact images, which is very useful for surface characterization, such as color, texture, and obvious spoilage signals. In this manner, the platform ensures that the system achieves good grading, which is required in the field and market situations, as the pork meat must be assessed reliably and quickly.

Table 4. Hardware Specification of the Smartphone in Testing

HARDWARE	SPECIFICATION
ANDROID OS	Android 14 Upside Down Cake
RAM	8 GB
PROCESSOR	Helio G100
CAMERA	50 MP

Data

This project consists of (4) meat cuts, those cuts are kasim, liempo, pork chop, and tenderloin, which can be classified: if it's fresh, or spoiled. The data is a formalized representation of facts, concepts, or instructions that can be processed by humans or machines, with meaning deeply tied to its specific context and purpose (B. K et al., 2020) and (Personally Acquired). The

standardized data sets consist of 6604 images with three classifications: fresh kasim (845), liempo (826), porkchop (810), tenderloin (804) and spoiled kasim (817), liempo (811), porkchop (868), tenderloin (823). Images were collected with phone cameras and captured from a 360-degree angular perspective.



Figure 5. Sample Data of Fresh (kasim, liempo, pork chop, tenderloin)



Figure 6. Sample Data of Spoiled (kasim, liempo, pork chop, tenderloin)

Model Development

The following procedures and methods are used to develop and finish the aforementioned mobile application in an orderly and systematic manner.

YOLOv8

As shown in figure 7, the step-by-step process for an input image to achieve accurate object detection. It begins with the backbone, where convolutional layers and C2f blocks extract both fine details and abstract

patterns, while the SPPF module enhances recognition by collecting features at multiple scales. The neck then fuses high-level and low-level features through upsampling and concatenation, creating a multi-scale feature map that strengthens detection across small, medium, and large objects. Finally, the head generates predictions by outputting bounding box coordinates, object classes, and confidence scores at three scales to ensure precise localization for objects of varying sizes. Supporting components, such as ConvModules, Darknet bottlenecks, and the SiLU activation function, improve learning efficiency, while advanced loss functions, such as Distribution Focal Loss for bounding box regression and Binary Cross-Entropy for classification, further enhance training and detection performance.

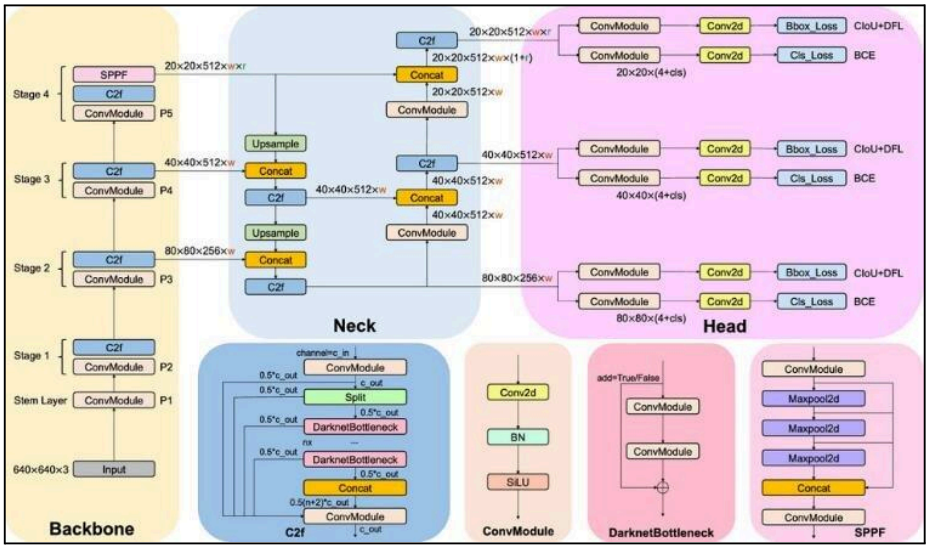


Figure 7. Diagram of YOLOv8

Model Development Process

As shown in Figure 8, the process of developing a pork freshness scanner begins with data collection, where raw information such as images, sensor readings, or text is gathered from various sources. In this case, thousands of pork images, including cuts such as pork chop, liempo, kasim,

and tenderloin, are captured under different lighting conditions, using HSV color conversion and CLAHE to ensure the model learns subtle differences in color, texture, and freshness.

Once collected, the data undergoes data preparation, which involves cleaning, organizing, and annotating the images. Each image is labeled according to freshness categories such as “fresh” or “spoiled.” The photos are then resized or normalized to maintain consistency, allowing the model to process them efficiently and learn patterns more effectively.

The next stage is model development, training, and validation. A suitable architecture, such as YOLOv8, is trained using the prepared and labeled dataset. The model learns to associate features with different freshness levels, and validation on unseen data ensures it generalizes well. Performance is assessed using metrics like accuracy, precision, recall, and loss.

Following training, the model is applied, with the trained YOLOv8 model deployed on a real-world platform, such as an Android mobile application. The app enables real-time predictions by processing live images captured by a smartphone's camera and classifying meat based on its freshness.

To sustain performance, the system undergoes model monitoring and updates. This involves tracking accuracy in real-world use and retraining the model with new data to adapt to changing conditions, such as variations in lighting, suppliers, or user environments.

Lastly, the process concludes with the outcome, in which the model provides predictions as classifications. For example, the application may display green for “fresh,” and red for “spoiled,” giving users clear, real-time feedback to support informed decisions about pork quality.

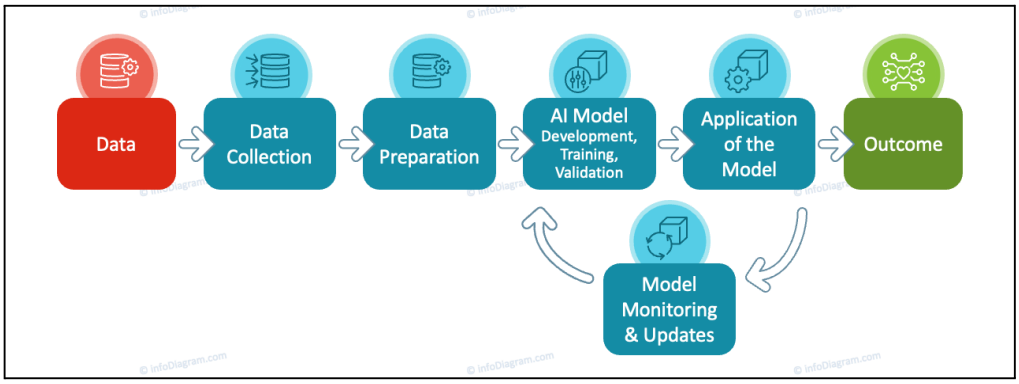


Figure 8. Diagram of Model Development

Evaluation Metrics

An evaluation metric is a table that summarizes how well the YOLOv8 model performs on a set of test data with known ground truth labels. It compares the actual (true) values to the predicted values produced by the model, helping to evaluate detection accuracy through metrics such as precision, recall, F1-score, and mean Average Precision (mAP).

As shown in Table 5, how well the model performs after testing. Precision means that when the model detects an object, it is usually correct, so there are only a few false alarms. Recall shows that the model can find most of the actual objects in the images and does not miss many of them. The F1-score summarizes both precision and recall, indicating that the model maintains a good balance between accuracy and completeness. The mAP@0.5 result suggests that the model works very well when a reasonable overlap between the predicted and actual bounding boxes is acceptable.

Meanwhile, the mAP@0.5:0.95 score reflects the model's performance under stricter conditions, where more precise box alignment is required, which naturally lowers the score. Overall, the table clearly shows that the model is reliable and performs well enough for real-world use.

Table 5. Metrics

Final Metrics	MEAN
Precision (mean)	0.8326
Recall (mean)	0.8709
F-1 score (mean)	0.8473
mAP@0.5	0.8721
mAP@0.5:0.95	0.5959

Unit Testing

As shown in Figure 9, the unit test life cycle commenced when the developer checked out the latest version of the code from the repository. Upon acquiring the code, the developer modified or polished the extant software components. After updates, the unit test was run to ensure the correct updates did not break. Whenever the developer found problems or defects during testing, the developer would fix these as they were found and rerun the unit tests to confirm the issues were resolved. Once all tests were successful, the code was examined for quality and coding guidelines. After the code was examined and validated, it was checked back into the respective repository, and the cycle was repeated to set the project up for later stages of development. This iterative process not only ensured that each component functioned correctly on its own but also contributed to the

robustness of the system as a whole. Focusing on the iterative cycle of testing, fixing, and validating code helped developers reduce the chances of creating errors in later stages of development. The cycle encouraged high code quality, strict standards, and a uniform process of development. It also made it much easier to integrate with other modules, helped the project's readiness for the next stage of the development process, and kept the software robust, maintainable, and ready for deployment.

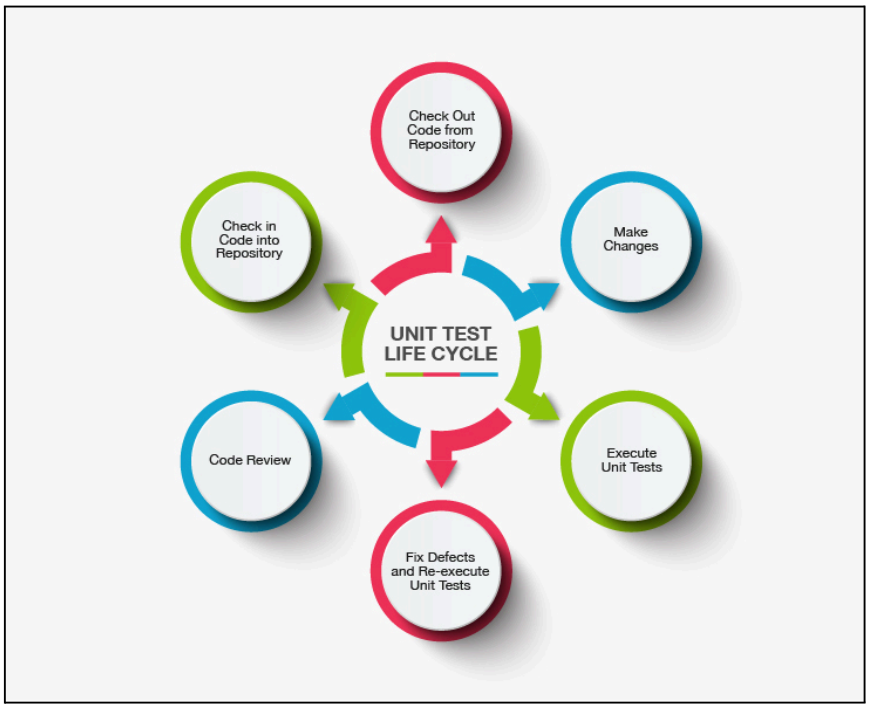


Figure 9. Unit Testing

Integration Testing

As shown in Figure 10, the developer checked out from the repository the latest version of the code. Subsequently, after the developer obtained the code, they edited or updated various existing components. Once the updates were complete, unit tests were run to ensure that the changes worked properly. Whenever problems or defects were found during testing, the developer fixed them and repeated the unit tests to check if the issues were

solved. After all tests had passed, the code was reviewed to ensure good quality and compliance with coding guidelines. Once the code was reviewed and approved, it was checked back into the repository, and the iterative process was repeated, preparing the project for subsequent stages of development. Such a cycle is useful for continual improvement and application development.

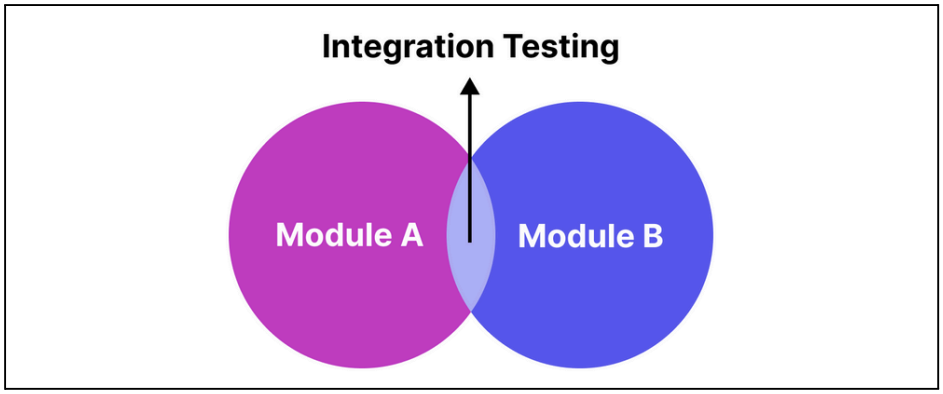


Figure 10. Integration

System Testing

As shown in Figure 11, the system testing was carried out to validate that the software meets the specification overall. Functional testing was conducted in Phase 1, where all features and functions were tested to verify that they performed as expected. After validating the system functionality, performance testing was performed to check speed, responsiveness, and stability under varying load levels. After performing performance tests, usability testing was conducted to prove that the system was easy to use, intuitive, and user-friendly. The final stage was security testing to identify potential vulnerabilities and to avoid any loss of data or unauthorized access. The system was extensively assessed and validated in this series of tests to ensure that it was ready to be deployed and used in the real world. This

structured testing sequence ensured that defects were identified early and corrected promptly. By verifying functionality, performance, usability, and security, the system proved its reliability and readiness for real deployment in real operating conditions. That way, this app would run correctly, remain stable, and deliver consistent results to the target user who uses it. This structured testing sequence ensured that defects were identified.

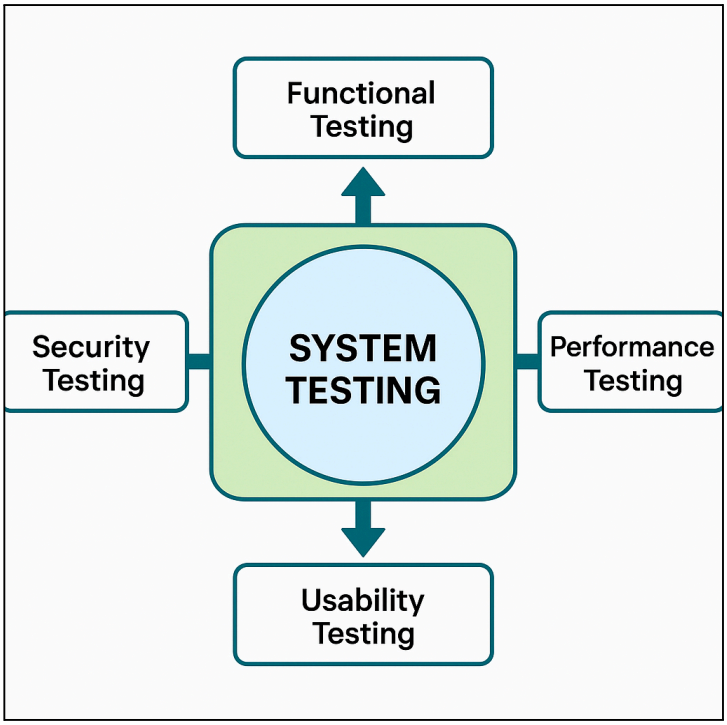


Figure 11. System Testing

Acceptance Testing

As shown in Figure 12, the acceptance testing involved a process to ensure the product's functioning met business requirements and was ready to be released. The first step was User Acceptance Testing (UAT), as the participants were the end-users who interacted with the software to ensure it worked and complemented a real-life workflow. Following UAT, the process proceeded to Operational Acceptance Testing (OAT) to verify whether, in its

real-world operational environment, the software was functioning properly. During this phase, backup procedures, maintenance routines, and overall system performance were assessed. This was an indication that the user and operational requirements were fully met and readiness for release was affirmed. This acceptance testing stage ensured the system was capable of functioning as expected, as required by actual user activities and operational demands. The feedback provided on the UAT and OAT served as assurance that the application behaved as intended, its daily operations were stable, and the expected state with which it would be deployed in general operation, with all outlined requirements, was met. Furthermore, this allowed for the recording and the correction of any minor issues or gaps that emerged from testing.

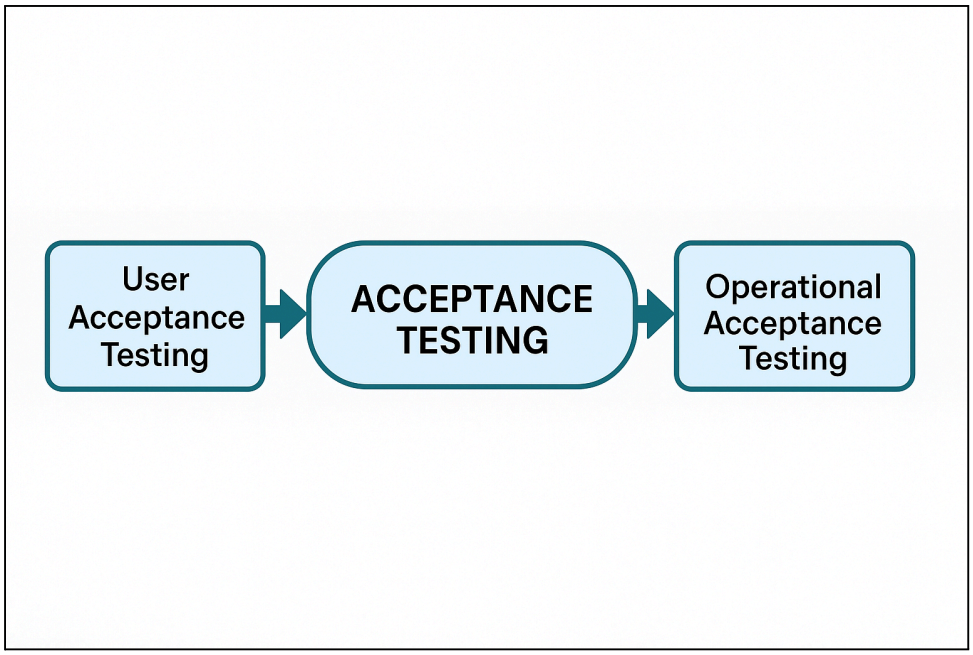


Figure 12. Acceptance Testing

Evaluation Procedure

As shown in Figure 13, Software product quality was assessed through the ways software carried out its intended functions, used resources

efficiently, and maintained compatibility with other systems. It also assessed user interaction, reliability, and security to ensure a smooth, secure, and trustworthy product. The evaluation also looked at the software's ability to be maintained, adapted, and safely integrated into different environments. This detailed evaluation was valuable in confirming that the software was functional, efficient, secure, reliable, user-friendly, and sustainable. This evaluation verified that the tool satisfied established quality standards and was still workable in practical operation. By assessing performance, usability, security, and maintainability simultaneously, the assessment validated that the software was able to sustain its planned behaviour over time. It also demonstrated that the system was responsive to new updates, could work consistently in different scenarios, and could continue to generate reliable outputs for the user as well as for the developer.

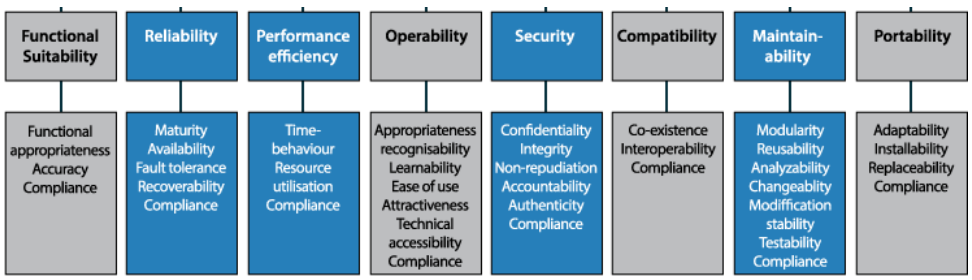


Figure 13. ISO 25010 Software Product Quality Characteristics

The Population of the Study

The study population included residents of Calamba City, Laguna, who were consumers who regularly purchased pork meat at the public market and vendors who sold it in this study. An information technology expert was also selected in the research to assess and test the application.

Sampling Design

The purposive sampling strategy (judgmental, selective, or subjective sampling) was employed as a nonprobability sampling approach with respondents selected based on their relevance and their perspective on the research. In total A total of 104 respondents were involved in the study. Three (3) IT experts assessed the system's technical acceptability, one (1) meat expert determined domain relevance, and one hundred (100) participants assessed the system using the Technology Acceptance Model (TAM).

Data Collection Instrument

The researchers also obtained approval from consumers and vendors to collect data through a consent letter. As part of the data collection process, face-to-face interviews were conducted to secure consent for accessing the dataset from public markets in Calamba, Laguna. Moreover, one meat expert, Mrs. Cristina Malabayabas, was also involved in the study as an experienced pork vendor, whose experience in handling and selling meat is approximately eight years. A printed questionnaire based on the Technology Acceptance Model (TAM) was used to measure user satisfaction with the application. IT experts and professionals completed the questionnaire manually, following the ISO 25010 software evaluation standard.

Statistical Treatment

Each response frequency was determined by the number of respondents who answered each item. The respondents' perspective was measured using a Likert scale. After data collection, the data were organized using a Microsoft Excel table, and the mean and standard deviation were

reported for analysis. Respondents were offered four choices for the assessment questions, ranked from highest to lowest in order of preference, as listed below: four (4) – Strongly Agree, three (3) – Agree, two (2) – Disagree, one (1) – Strongly Disagree. This ranking was critical to assess respondents' answers. The means and standard deviations were used to summarize the responses, and the scores were averaged to obtain the composite mean to represent general interpretation by the respondents.

As shown in Table 5, the Likert scale was used in the survey to gauge user satisfaction with the application.

The respondents' evaluations are measured using a percentage technique. The formula is as follows:

Where:

$$\frac{\text{Highest Score - Lowest Score}}{\text{Highest Score}} \qquad \begin{matrix} 4 - 1 = 3 \\ 3 / 4 = 0.75 \end{matrix}$$

Table 6. Likert Scale Method

RATING	INTERPRETATION	RANGE
4	Strongly Agree	3.25 - 4.00
3	Agree	2.50 - 3.24
2	Disagree	1.75 - 2.49
1	Strongly Disagree	1.0 - 1.74

CHAPTER 4

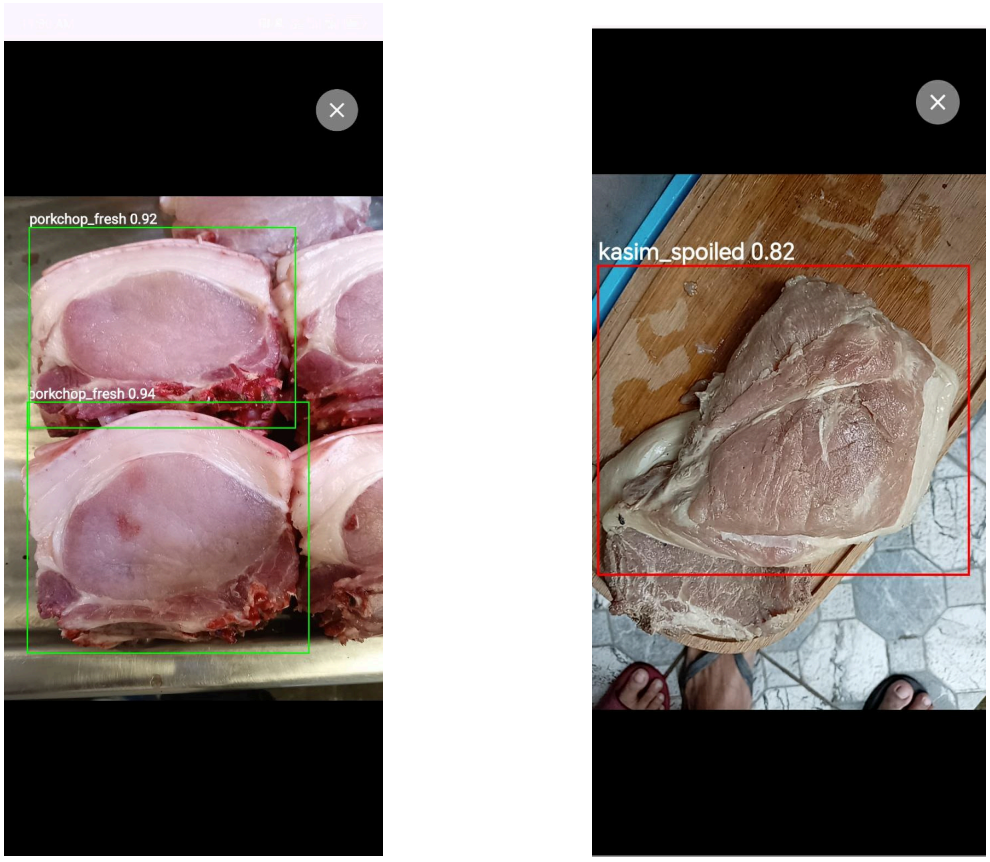


Figure 14. Detection Result

As shown in Figure 15, a system capable of multiple object detection with classification for identifying the cuts and freshness of pork meat, the study built the proposed model shown in Figures 16 to 21 summarized the overall system workflow, showing how the application processed uploaded or captured images, detected multiple pork cuts within a single frame, and generated freshness classifications with corresponding confidence scores. The figures also demonstrated that each detected object was independently analyzed and accurately labeled throughout the process. Also showed that system accuracy was maintained even in the presence of multiple pork cuts and that each bounding box, label, and confidence score was correctly associated with the object that was identified.

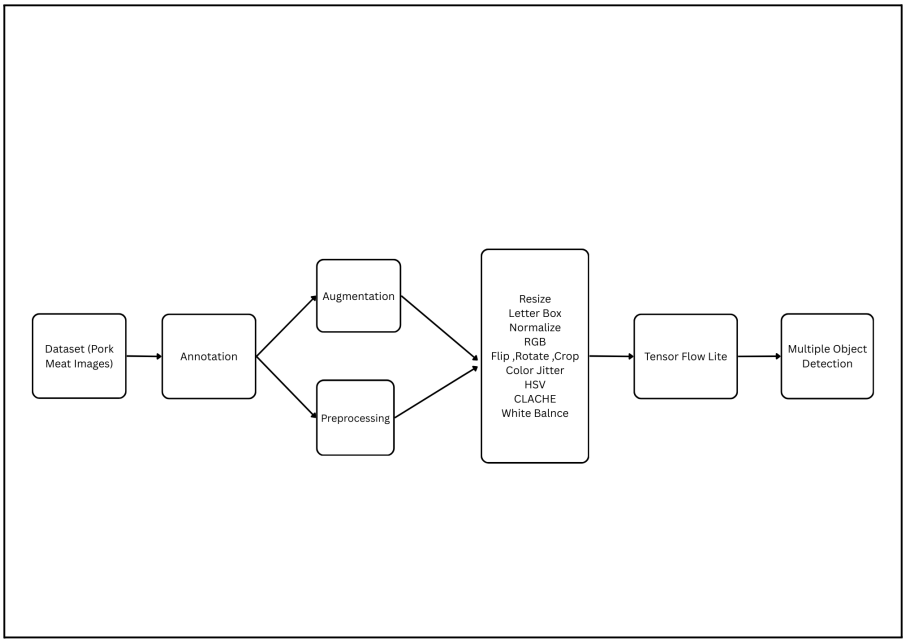


Figure 15. Object Detection with Classification Architecture

As shown in Figure 16, the annotation process started by opening an image from the dataset. A bounding box was drawn around the meat, which would indicate to the AI which object was object to focus on. Then the box was reshaped to fit the meat properly, with no unnecessary background. Then, when the proper adjustment was applied, the appropriate name for the meat cut was selected from the class list. After this, the annotation was saved and carried out on the next image until all images in the dataset were labeled. Once the bounding box was properly aligned, the correct label for the given pork meat cut was selected from the predefined class list. Doing so helped us find the respective class corresponding to every single area of annotation, which would provide some guidance during the learning and classification phases. After the annotation was finished, it was saved, and the same method was carried out for the other pictures in the dataset until all images were fully annotated.

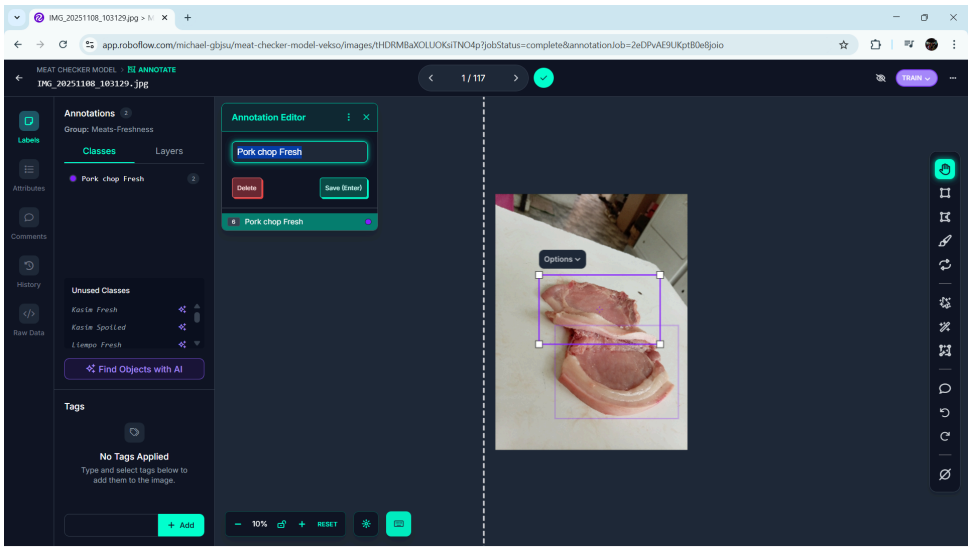


Figure 16. Annotation Process using Roboflow

As shown in Figure 17, Augmentation started by applying variations to the images, such as adjusting brightness, slightly rotating, or flipping them. This way, it was able to get used to seeing the meat in different contexts and even recognize it after it actually looks different in an everyday environment. This process increased the diversity of the training dataset and helped prevent overfitting during model training. By exposing the model to varied visual conditions, augmentation improved its ability to generalize and maintain stable detection performance when applied to real-world images captured under different lighting, angles, and orientations. As a result, the YOLOv8 model became more robust and adaptable to practical scanning situations. The augmented images allowed the system to better handle inconsistencies commonly found in market environments, such as uneven lighting or slight camera movement. This contributed to more reliable freshness detection and consistent classification results during actual application use.

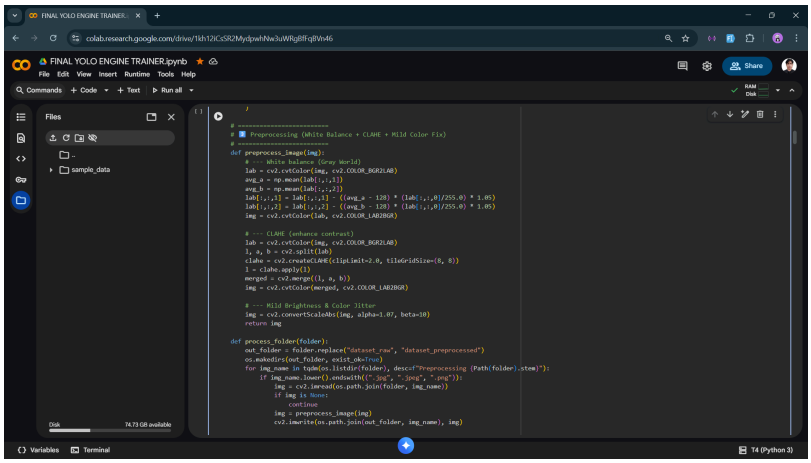


Figure 17. Dataset Augmentation using YOLOv8

As shown in Figure 18, the preprocessing process started with cleaning and enhancing each image to better understand this AI. It adjusted color tones, enhanced contrast to highlight specific details with better focus, and balanced brightness and colors. Each image that was processed was scripted, and the improved images were saved in the proper folder, ready for training. The data was made consistent, precise, and unambiguous before being fed into the model in the step. As a result, preprocessing rendered features more visible and, in turn, allowed more precise learning in training by standardizing image quality, reducing visual noise, or similar aspects. In other words, the model could find more information, useful patterns in pork meat images, also for more accurate detection performance, and classify correctly. Preprocessing contributed to reducing variations caused by different capture devices and environments. With the consistent properties of the images in the dataset, the training of the model was stable and efficient. This preparation phase accelerated convergence during training and ensured uniformity in performance upon the system being developed for real-world pork meat scanning.

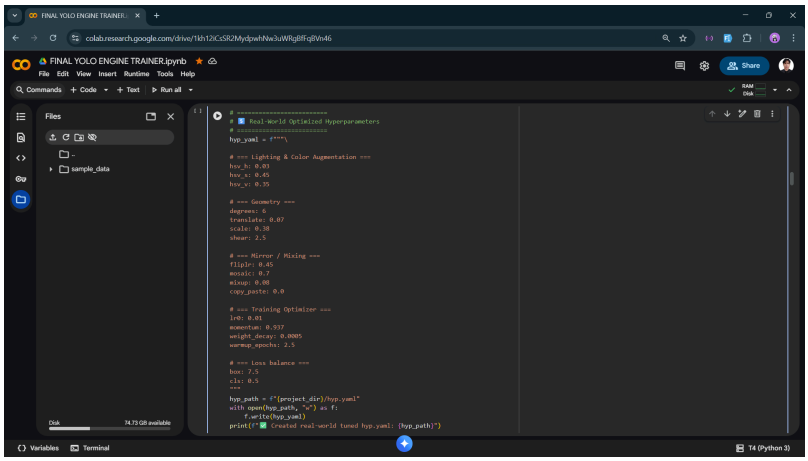


Figure 18. Preprocessing dataset using YOLOv8

As shown in Figure 19, the code imported a YOLOv8 model converted to TensorFlow Lite, activated it with the TFLite interpreter, and verified the expected shape and type of data entered by the model to determine whether it was suitable for inference. It scanned the whole image for regions with consistent shapes with known objects. If a potential object was identified, a bounding box would be drawn around that portion as well. The model generated prediction scores that indicated the probability of every detected object belonging to a given freshness category after inference. We then filtered our results according to confidence thresholds to reduce false detections. The remaining valid detections were prepared for display to provide meaningful viewing to the user through easy retrieval in bounding boxes and classification.

```
import tensorflow as tf

model_path = "/content/nano_model.tflite"
# Load your TFLite model
interpreter = tf.lite.Interpreter(model_path)
interpreter.allocate_tensors()

# Get input details
input_details = interpreter.get_input_details()
print(input_details)

[{'name': 'images', 'index': 0, 'shape': array([  1, 416, 416,   3], dtype=int32), 'shape_signature': array([  1, 416, 416,   3], dt
```

Figure 19. Converting the YOLOv8 model into TensorFlow Lite

As shown in Figure 20, the entire image was scanned by the model to identify regions that matched the learned features and shapes of known pork meat objects. When a potential object was detected, a bounding box was generated around the corresponding area to isolate it for further analysis. Each detected region was then evaluated by the model and classified into its respective category, such as “kasim_fresh” or “porkchop_fresh,” based on the patterns and visual characteristics extracted during training. This classification relied on both the spatial features and texture cues learned by the model, which allowed it to distinguish between subtle differences in meat cuts and freshness levels.

After classification, a confidence score was assigned to each prediction, indicating the degree of certainty associated with the model’s output. This scoring provided a quantifiable measure of reliability, which enabled users to interpret the predictions with greater trust. The system also prioritized high-confidence detections when making automated decisions, ensuring that less certain predictions were flagged for manual review.

Overall, this process illustrated the effectiveness of the integrated detection pipeline in performing object localization, classification, and confidence scoring. By combining these steps into a seamless workflow, the system contributed significantly to automated pork meat freshness assessment, offering fast, accurate, and reliable results that could be leveraged in retail, processing, and quality control environments. The approach demonstrated scalability, allowing the model to recognize more

meat types or quality attributes with minimal retraining, highlighting its potential for broader food inspection applications.



Figure 20. Multiple Object Detection with Classification

To implement the YOLOv8 Android mobile application interface and integrate the YOLOv8 model to enable a real-time pork meat scanner and visual freshness classification for consumers and vendors.

As shown in Figure 21, the Home Screen of the MeatChecker application was displayed. The screen also provided dropdown sections for pork cuts, beside the “Scan Now” button. Once the user extended a dropdown, it showed a quick snippet of the exact pork cut, along with the part of the pig, the optimal cooking method, and a fun fact. This enabled the user

to learn more about the characteristics of pork meat before any scan. When ready, the user clicked on “Scan Now,” which pushed him to start up his camera for real-time scanning using the YOLOv8 model. The layout guaranteed an informative and user-friendly interface that would be appropriate for both consumers and vendors.

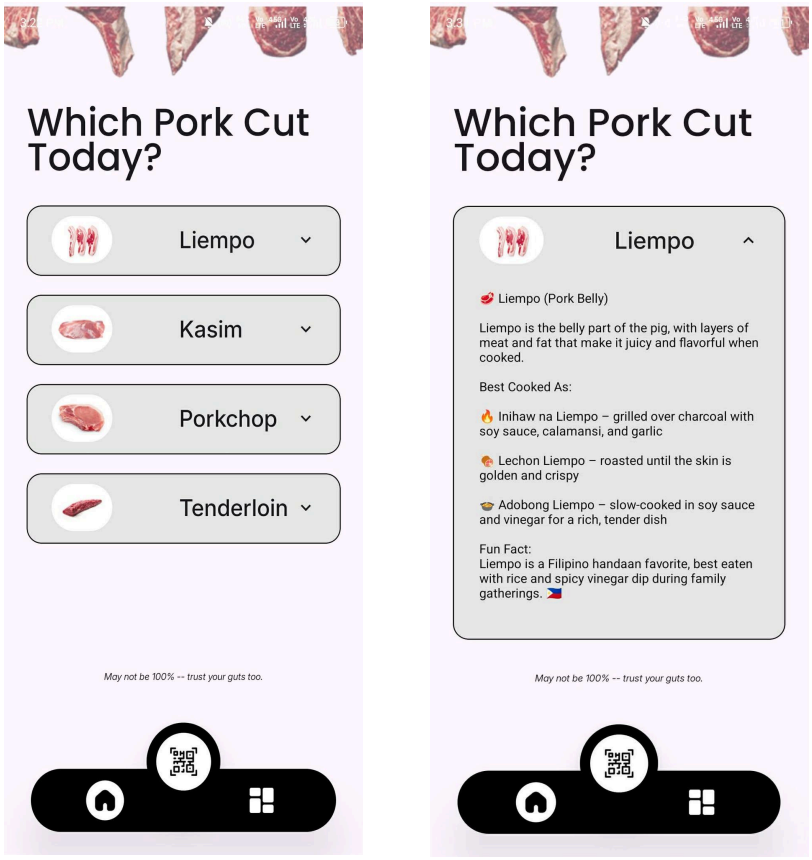


Figure 21. Home Screen

As shown in Figure 22, the Camera Scanning Screen appeared. When the device tapped “Scan Now,” it opened the camera window and showed an actual preview of what it’d seen live. When the user snapped any image of pork, the image was sent to the YOLOv8 model, which analyzed and ranked the meat’s freshness. This was also the screen where the detection process officially began.

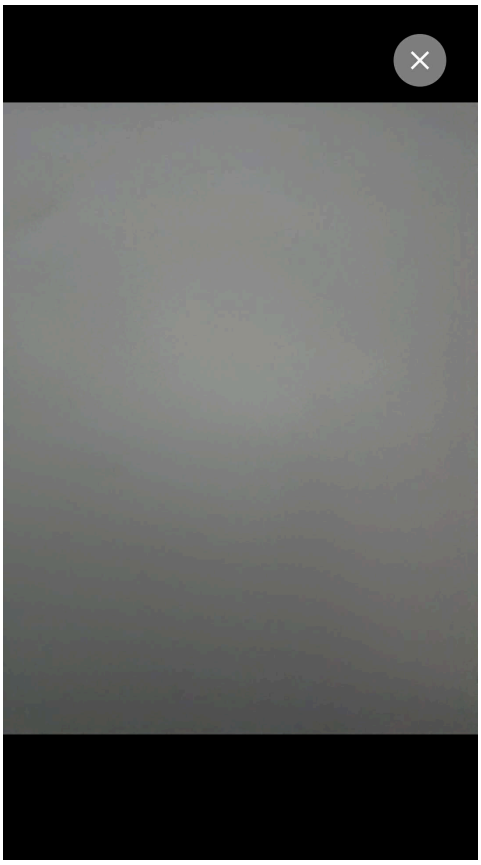


Figure 22. Camera Scanning Screen

As shown in Figure 23, the detection results after scanning were displayed by the system. It identified the pork cut, presented freshness classification, and the confidence percentage yielded by the YOLOv8 model. Higher confidence scores (80–100%) implied more certainty in the model, and lower values indicated more doubt as a consequence of variations in lighting, angle, or varying degrees of occlusion in the meat. Before saving, the user reviewed the image and read some recommendations for cooking the item. The report could then be canceled or saved by the user. At this stage, users had well-constructed and helpful facts before reaching an action decision. It facilitated users' easy evaluation of results by combining visual cues, confidence percentages, and concise recommendations. Cancel or save the

scan; this way, users have control over which records were retained and can refer back to them as needed. The display of confidence showed users how reliable each detection was. Such transparency resulted in trust for the system from the user, and for cautious interpretation of results, especially when scanning conditions were not ideal. Overall, the results display enhanced usability and emphasize the practical usefulness of the MeatChecker application in real-world scenarios.

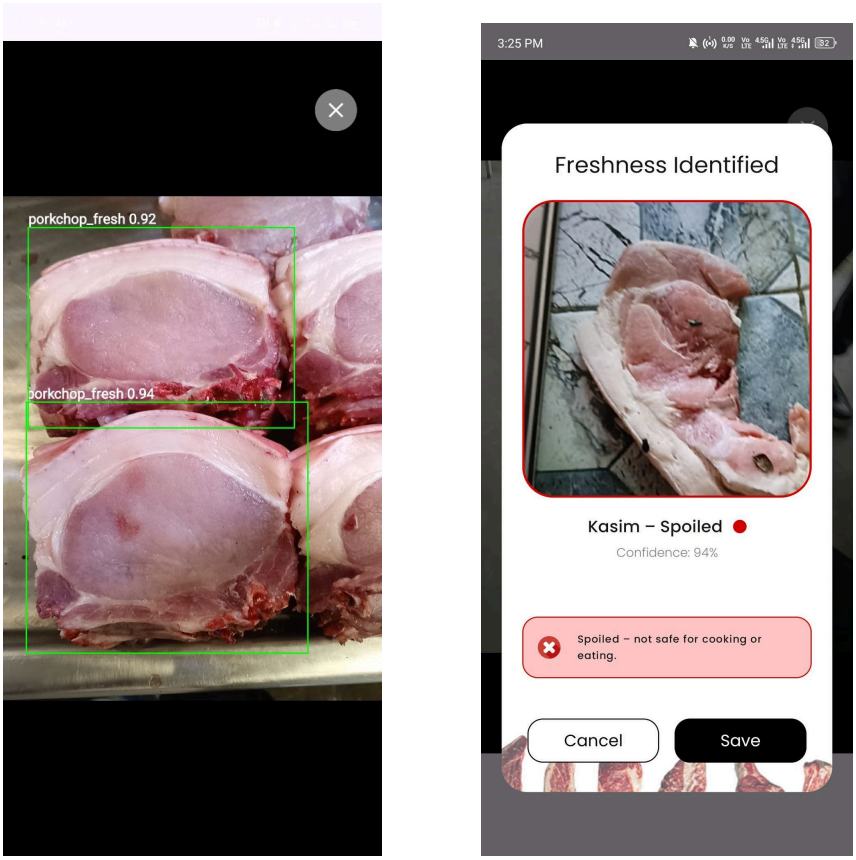


Figure 23. Detection Result and Preview Screen

As shown in Figure 24, the Dashboard screen. This page serves as the report section of the application, where users can view a summary of their scanned pork results. The dashboard displays daily scanned data, showing the number of pork samples classified as Fresh and Spoiled based on the

YOLOv8 detection results. Each category includes a brief description of the meat characteristics and the number of times that classification occurred on that day. This allows users to monitor and review their scanning activity easily. This screen provided a consolidated summary of the user's scanning logs and detection results in a simpler, readable way. The dashboard helped users to quickly identify pork freshness patterns without checking individual scans due to the daily outputs being summarized. The clear distinction of Fresh and Spoiled counts made the information easy to comprehend. The dashboard served as an actionable tracking element enabling users to evaluate the scan frequency and result consistency over time. It served as a feature to view and compare daily observations in addition to organizing data, to allow reporting purposes, and enhance the usefulness of the app.

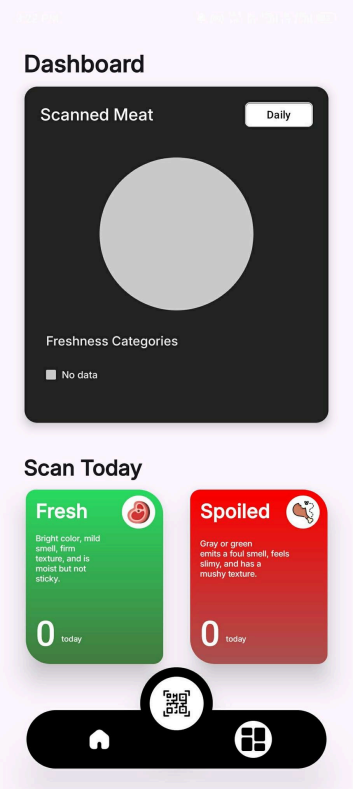


Figure 24. Dashboard (Reports Screen)

Test and Evaluate the Software using ISO 25010

According to the statistics in Table 7, the lowest beta test mean result, according to the IT technicians, was 3.00, also meaning a verbal response of “Agree,” and the maximum was 4.00, which means “Strongly Agree.” With an overall mean for the level of agreement of 3.69, IT professionals seemed to agree to some extent. In all such evidence, the system was accepted as satisfactory; satisfactory in terms of test acceptance and the level of acceptability at the beta stage.

The findings also showed that the experts’ opinions on the features, usability, and functionality of the system were quite positive. The range at which minimum and maximum scores were derived from slightly different opinions between participants reflected some slight variability in perception, which may be due to personal preference, previous experience in using similar software, other experience, or expectations about their understanding of certain features. This diversity was also prevalent during the beta test process, where users emphasized different aspects of the system depending on their role or expertise.

Additionally, the steady positivity of responses suggested that the system met the intended goals of usability and responsiveness (including responsiveness and accuracy of output). The beta testing results finally showed that the system met professional standards, received IT endorsement, and was also ready to deploy to others. The results also proved insightful in suggesting future improvements that would have increased its user satisfaction and operational efficiency.

Table 7. Results of Beta Testing answered by the IT Professional

INDICATORS	MEAN	STANDARD DEVIATION	INTERPRETATION
Functional Completeness	4	0	Strongly Agree
Functional Correctness	3.33	±0.58	Strongly Agree
Functional Appropriateness	3.33	±0.58	Strongly Agree
Time Behaviour	4	0	Strongly Agree
Resource Utilization	4	0	Strongly Agree
Capacity	3.67	±0.58	Strongly Agree
Co-existence	3.67	±0.58	Strongly Agree
Interoperability	3.67	±0.58	Strongly Agree
Appropriateness Recognizability	3.67	±0.58	Strongly Agree
Learnability	4	0	Strongly Agree
Operability	4	0	Strongly Agree
User Engagement	3.67	±0.58	Strongly Agree
Inclusivity / Accessibility	4	0	Strongly Agree
Availability	4	0	Strongly Agree
Fault Tolerance	3	0	Agree
Recoverability	3.67	±0.58	Strongly Agree
Consistency	3	0	Agree
Confidentiality	4	0	Strongly Agree
Authenticity	4	0	Strongly Agree
Resistance	4	0	Strongly Agree
Analysability	3.33	±0.58	Strongly Agree
Modifiability	3.33	±0.58	Strongly Agree
Adaptability	3.67	±0.58	Strongly Agree
Scalability	3.33	±1.15	Strongly Agree
Installability	4	0	Strongly Agree
Replaceability	3.67	±0.58	Strongly Agree
Overall	3.69		Strongly Agree

Legend: 4 – Strongly Agree 3 – Agree 2 – Strongly Disagree 1 – Disagree

Test and Evaluation Performed using TAM

The results are collected from the respondents, who are the consumers and vendors that would use the application.

As shown in Table 8, for a mean of 3.60, “Strongly Agree,” the users thought the system had greatly improved efficiencies in pork meat grading by freshness and quality. Users also indicated that, as a direct result, the system had provided accurate, up-to-the-moment grading results, which facilitated more prudent meat selection (An average of 3.06, “Agree”). Moreover, pork quality assessment also correlated with increased productivity and consistency found in pork meat grading (M = 3.19).

Results were even more promising, in that the system had played a role in the recognition of meat quality issues before the sale or consumption of the product, also achieving an average score of 3.19. Users also reported a useful system in the process of maintaining quality pork meat for vendors and consumers, averaging 3.17. For the majority of the answers, respondents mostly “Agreed” to “Strongly Agreed” on the operational, dependable, and helpful characteristics of the system in providing the support to the correct and regular pork meat grading. The results indicated that the approach had effectively solved technical issues encountered when manually grading the meat and decreased human-related error, and ensured consistency in evaluation across different evaluators. Users were also reported to have seen that the system supported informed decision-making because real-time

grading let vendors quickly identify issues and respond to them to ensure quality.

Table 8. Level of Perceived Usefulness

STATEMENTS	MEAN	STANDARD DEVIATION	INTERPRETATION
The system improves the efficiency of grading pork meat based on freshness and quality.	3.60	±0.85	Strongly Agree
The system provides accurate and real-time grading results that support better decision-making in meat selection.	3.08	±0.80	Agree
Using this system increases productivity and ensures consistency in pork meat quality assessment.	3.19	±0.77	Agree
The system helps identify meat quality issues before the product is sold or consumed.	3.19	±0.77	Agree
I find the system valuable in ensuring high-quality pork meat for vendors and consumers.	3.17	±0.73	Agree
Overall	3.17		Agree

Legend: 4 – Strongly Agree 3 – Agree 2 – Strongly Disagree 1 – Disagree

As shown in Table 9, users perceive the mobile application interface as user-friendly and easy to navigate, with a mean of 3.12 (“Agree”). Respondents also agree that the application is easy to learn and that the grading results are easy to understand, as reflected in a mean of 3.04.

Furthermore, the system is perceived to require minimal effort or technical knowledge to operate, as shown by a mean of 3.14. The application is also considered convenient for capturing pork images and viewing grading results, as evidenced by a mean of 3.07. Lastly, users express comfort with the overall setup and functionality of the MeatChecker application, as evidenced by a mean score of 3.06. Overall, the results reflected a positive perception of usability, showing that users had found the system accessible, manageable, and easy to operate, which suggested that the design had effectively supported user interaction and had minimized learning difficulties.

Table 9. Level of Perceived Ease-of-Use

STATEMENT	MEAN	STANDARD DEVIATION	INTERPRETATION
The system interface is user-friendly and easy to navigate.	3.12	±0.84	Agree
I find it easy to learn how to use the application and understand the grading results.	3.04	±0.75	Agree
Using the system requires minimal effort or technical knowledge.	3.14	±0.84	Agree
The application makes it convenient to capture pork images and view grading reports.	3.07	±0.74	Agree
I am comfortable with the overall setup and functionality of the MeatChecker application.	3.06	±0.78	Agree
Overall	3.09		

Legend: 4 – Strongly Agree 3 – Agree 2 – Strongly Disagree 1 – Disagree

Result of Software Testing

The mobile application user interface was user-friendly and easy to navigate, with an average of 3.12 (“Agree”). They were also consistent in saying that the application was easy to learn and that grading results were easy to understand, which reflected a mean of 3.04. In addition, the system was viewed as requiring little effort or technical knowledge to use, with a mean of 3.14. The app was also deemed useful for taking pictures of pork and for comparing grading results, reflected in the mean score of 3.07. At the end, users reported being somewhat comfortable with the MeatChecker app with respect to its setup and functionality, with a mean score of 3.06. In general, the findings showed a satisfactory perception of usability, meaning that users perceived their system to be accessible, manageable, and user-friendly.

Unit Testing Result

As shown in Table 10, test the units of the system or individual system, and for each system, there was one key task performed with that unit. The related sub-systems inside the modules were all tested for quality. There, any inconsistencies were discovered and corrected to result in a satisfactory system. After the many devices were implemented, the team reviewed them to find out whether the modules had managed well not only to the reliability of the system but also at a high level. Minor issues identified in the test cycle were fixed promptly, which helped to increase the system stability and instilled confidence in users. Overall, the evaluation showed that the built system

worked in real-life scenarios and was sufficiently efficient with the support of its design.

Table 10. Unit Testing for Home Screen Module

Test Case	Expected Output	Actual Output	Remarks
Home Screen	The application would show the home screen interface with the “Scan Now” button and dropdown sections for pork cuts.	The application successfully showed the home screen interface with all buttons and dropdowns visible.	Passed
Dropdown Functionality	The application would show pork cut information, including its location in the pig, how best to cook as, and a fun fact when a dropdown is tapped.	The application successfully showed the correct and complete information for each pork cut.	Passed
Navigation to Scanning	The application would show the camera scanning screen for real-time detection when the user taps “Scan Now.”	The application successfully showed the Camera Scanning Screen upon pressing “Scan Now.”	Passed

As shown in Table 11, the Home Screen (or Dashboard module) passed unit testing. There was no problem with how the appropriate application appeared with the home interface; they showed the “Scan Now” button and dropdown sections for pork cuts as expected. The dropdown worked well, and the location of the pork cut, recommended cooking method, and interesting facts were inserted when the button was pressed. For this, too, the click to the Camera Scanning Screen went well after hitting the “Scan

Now” button. Finally, all the tests of this module proved to be successful, according to the comments, as it had been confirmed that the Home Screen worked as designed, too.

Table 11. Unit Testing for Camera Scanning Screen Module

Test Case	Expected Output	Actual Output	Remarks
Camera Activation	The application would show the live camera preview after the user taps “Scan Now.”	The application successfully showed the live camera preview.	Passed
Real-Time Scanning	The application would show the real-time detection process as the YOLOv8 model analyzes the pork meat in the camera feed.	The application successfully showed real-time detection results during scanning.	Passed
Model Operation	The application would show the analysis results once the YOLOv8 model detects a valid pork image.	The application successfully showed the detection results after model analysis.	Passed

As shown in Table 12, the detection result and the preview screen module completed one successful unit test. Correct detection results were displayed by the application (name of pork cut, freshness classification, confidence level, bounding box indicators) that indicated where the detection region was. The preview of the image had looked correct, and users had been able to see both the scanned image and its visual markers. It was concluded that all the tests of this module had been completed, and the comments confirmed that the detection visualization and preview functionality had

worked correctly. Furthermore, the reliable operation of the save and cancel functions suggested that the module supported efficient workflow management and minimized potential user errors.

Table 12. Unit Testing for Detection Result and Preview Screen Module

Test Case	Expected Output	Actual Output	Remarks
Result Display	The application would show the pork cut, freshness classification, confidence percentage, and bounding box indicators during scanning.	The application successfully showed the correct detection results and bounding box indicators.	Passed
Image Preview	The application would show the scanned image along with bounding boxes and detection details when tapping the area or pork cut on the live screen for user review.	The application successfully showed the image preview with bounding boxes and related details.	Passed
Save/Cancel Function	The application would show save and cancel options that allow the user to confirm or discard the result.	The application successfully showed functional save and cancel buttons.	Passed

As shown in Table 13, the Unit test passed successfully. As we expected to see scanned pork classifications on the application with their respective details for each category. For example, the data visualization feature performed as expected for counts and descriptions per class per day and week. In addition, the navigation test was completed, allowing users to return to the Home Screen without issues. As per the comments, all tests for

this module were passed, a key aspect of testing a reporting and tracking system.

Table 13. Unit Testing for Dashboard (Reports Screen) Module

Test Case	Expected Output	Actual Output	Remarks
Report Summary	The application would show a summary of scanned pork results categorized as Fresh and Spoiled.	The application successfully showed the summarized report with accurate data.	Passed
Data Visualization	The application would show detailed counts and descriptions for each classification per day or week.	The application successfully showed the correct data visualization.	Passed
Navigation	The application would show the home screen when the user chooses to return from the dashboard.	The application successfully showed the home screen after navigating back.	Passed

Integration Testing Result

Integration Testing demands the integration of all modules that contribute to the system's functionality. Connecting the Home Screen module, Camera Scanning Screen module, Detection Result and Preview Screen module, and Reports / Dashboard Screen module would boost the overall performance of the working system.

As shown in Table 14, Integration testing involved checking all of the modules that were working together as part of the system to ensure that they actually worked. The addition of the Home Screen module, Camera Scanning

Screen module, Detection Result and Preview Screen module, and Reports/Dashboard Screen module has successfully contributed to the improvement in the overall performance of the MeatChecker app.

Table 14. Integration Testing of the System

Test Case	Expected Output	Actual Output	Remarks
Home Screen Module	The application can show the Camera Scanning Screen when the user taps “Scan Now” on the Home Screen.	The application can successfully show the Camera Scanning Screen after tapping “Scan Now.”	Passed
Camera Scanning Screen Module	The application can show the detection results, including the pork cut, freshness classification, confidence percentage, and bounding box indicators after real-time scanning.	The application can successfully show the detection results with all relevant indicators after scanning.	Passed
Detection Result and Preview Screen Module	The application can successfully show the saved detection results in the Dashboard after the user chooses to save the result.	The application can successfully show the saved detection results in the Dashboard with accurate data.	Passed
Reports / Dashboard Screen Module	The application can successfully show the Home Screen when the user chooses to return or navigate back from the Dashboard.	The application can successfully show the Home Screen after navigation from the Dashboard.	Passed

System Testing Result

As shown in Table 15, the system acceptance testing criteria of functionality, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability were assigned “Strongly Agree”. Acceptance test results show that, according to the selected respondent category, the application users were using the application. The acceptance test had also enabled the developer to have the end-users involved in the testing process and give ratings, which allowed the developer to realize the functionality required for each of the functionalities provided in the software being tested.

The results indicated that the system accomplished technical and user-centered requirements which made it ready to deploy. The ongoing positive feedback showed that the software had solved real-world use cases well and that it was aiding the users to do their jobs quickly.

Furthermore, the acceptance tests showed that the system structure & functionalities also matched with what the user's expectation, which prevented the user from making mistakes during actual use.

Moreover, the positive acceptance results showed that the system had reached a high reliability and stability, which had increased user confidence during regular use.

The feedback also pointed out that the software was intuitive and easy to use, thus minimizing the learning curve for new users. Acceptance testing had shown the developers how to take actions to enhance some functionalities to allow for continual improvement in future iterations.

Table 15. System Testing Result

Requirement	Process	Expected Result	Actual Result	Remarks
Real-Time Pork Scanning	Capture live pork images using the device's camera and process them through the YOLOv8 model.	The application can successfully analyze the image and display the detection result within 1 second.	0.9 seconds	Passed
Freshness Classification	Analyze the pork image and determine the freshness level	The application can successfully classify the pork as Fresh and Spoiled based on the model output.	1 seconds	Passed
Report Generation	Save scanned results and display them in the Dashboard	The application can successfully generate reports and display saved scans in the Dashboard.	0.9 seconds	Passed
Information Module	The application can successfully display pork cut information through a dropdown selection.	The application can successfully show pork cut details such as the location of the part, how best to cook as, and a fun fact.	0.5 seconds	Passed

Acceptance Testing Result

The results are the respondents' responses that are collected, who are the IT experts who would use and evaluate the system. These acceptance testing results show whether the application meets the acceptance criteria.

As shown in Table 16, the system's acceptance testing focused on the characteristics of functionality, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability, all of which received interpretations of "Strongly Agree." Acceptance testing allowed the developers to involve end users in the testing process and gather their feedback through ratings, providing valuable insights into the effectiveness and suitability of each function in the tested software.

Table 16. Acceptance Testing of the System

Characteristics	Mean	Interpretation
Functional Sustainability	3.55	Strongly Agree
Performance Efficiency	3.89	Strongly Agree
Compatibility	3.67	Strongly Agree
Usability	3.84	Strongly Agree
Reliability	3.42	Strongly Agree
Security	4	Strongly Agree
Maintability	3.33	Strongly Agree
Portability	3.67	Strongly Agree
Overall	3.67	Strongly Agree

Legend: 4 – Strongly Agree 3 – Agree 2 – Strongly Disagree 1 – Disagree

CHAPTER 5

Conclusion

MeatChecker: An Android Mobile Application for Grading Pork Meat, developed by applying the YOLOv8 Algorithm, examined pork meat pictures and graded them to help with freshness. Pig meat has some features that may be challenging to accurately assess by regular consumers or vendors, as freshness can only be judged based on subtle visual attributes such as color or appearance. Not everyone can tell with certainty whether pork meat is fresh or spoiled - and that can mean bad eating decisions or health woes. The user only had to do an image upload or take a pic of the pork meat, which was processed via the YOLOv8 model for the pork cut and freshness level detection. By analyzing its image, the application had successfully classified pork meat in terms of fresh or spoiled, all of which are classified by the dataset. Different kinds of pork were correctly detected, and fresh nature classifications were given, helping consumers, as well as vendors, in assessing the quality of this meat. The application system used an automated detection mechanism to obtain data about pork meat status and provided an easy, real-time estimation. According to the ISO 25010 software quality standard, the application was designed to satisfy user requirements for: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. The developed Android application was found to be very satisfactory by the researchers in accordance with the ISO 25010 software quality indicators. These results imply that MeatChecker has facilitated easier analysis of pork meat quality and has made decisions concerning handling or buying pork meat safer and more informed.

Recommendations

The recommendations below are based on the project findings. The following recommendations should be pursued to improve the application:

1. This study should include additional pork cuts and different types of meat or poultry to broaden the system's classification capabilities.
2. Future researchers may search for the latest version of yolo that can be used and applicable in Image processing and multiple object detection for a better experience.
3. The future researcher may use CNN for better classification of meat freshness.
4. The future researcher may add an upload function to the app and not only live scanning.
5. Future versions of MeatChecker can integrate simple IoT external sensors for better support in freshness detection.

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LIST OF APPENDICES

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APPENDIX A

Planning & Requirement Analysis Phase

Survey Questionnaire

TECHNOLOGY ACCEPTANCE MODEL QUESTIONNAIRE

Dear respondent,

We are students of the Laguna State Polytechnic University – Los Baños Campus under the College of Computer Studies conducting development on the project entitled **"MEATCHECKER: AN ANDROID-BASED MOBILE APPLICATION FOR PORK MEAT GRADING UTILIZING YOLOv8 ALGORITHM"** We would like to seek your opinion on the acceptability of the said project by scoring the statements below by checking the appropriate numbers, wherein 4 = Strongly Agree; 3 = Agree; 2 = Disagree; and 1 = Strongly Disagree. Rest assured that any information that you provide would be treated with utmost confidentiality (RA 10173. Data Privacy Act) and would be used for research purposes only. Thank you for your cooperation.

Proponents:
Delo Santos, Michael M.
Sambitan, Kyla Marie T.
Valera, Jay vee O.

NAME OF RESPONDENT:
_____

Perceived Usefulness	**4**	**3**	**2**	**1**
1. The system improves the efficiency of grading pork meat based on freshness and quality. *Pinapahusay ng sistema ang kahusayan sa pag-grado ng karne ng baboy batay sa kasariwaan at kalidad.*				
2. The system provides accurate and real-time grading results that support better decision-making in meat selection. *Nagbibigay ang sistema ng tumpak at real-time na resulta ng pag-grado na nakatutulong sa mas mabuting pagpapasya sa pagpili ng karne.*				
3. Using this system increases productivity and ensures consistency in pork meat quality assessment. *Ang paggamit ng sistemang ito ay nagpapataas ng produktibidad at katiyakan sa pagtatasa ng kalidad ng karne ng baboy.*				

4. The system helps identify meat quality issues before the product is sold or consumed. *Nakatutulong ang sistema na matukoy ang mga problema sa kalidad ng karne bago ito ipagbili o kainin.*				
5. I find the system valuable in ensuring high-quality pork meat for vendors and customers. *Mahalaga sa akin ang sistemang ito upang matiyak ang mataas na kalidad ng karne ng baboy para sa mga tindero at mamimili.*				
Perceived Ease of Use				
1. The system interface is user-friendly and easy to navigate. *Madaling gamitin at i-navigate ang interface ng sistema.*				
2. I find it easy to learn how to use the application and understand the grading results. *Madali kong natutunan kung paano gamitin ang aplikasyon at intindihin ang mga resulta ng pag-grado.*				
3. Using the system requires minimal effort or technical knowledge. *Ang paggamit ng sistema ay nangangailangan lamang ng kaunting pagsisikap o teknikal na kaalaman.*				
4. The application makes it convenient to capture pork images and view grading reports. *Ginagawang maginhawa ng aplikasyon ang pagkuha ng larawan ng karne at pagtingin ng mga ulat ng pag-grado.*				
5. I am comfortable with the overall setup and functionality of the MeatChecker application. *Komportable ako sa kabuuang setup at mga tungkulin ng MeatChecker application.*				
Attitude (Towards Using the System)				
1. I feel positive about using the system for pork meat grading. *Mayroon akong positibong pananaw sa paggamit ng sistema para sa pag-grado ng karne ng baboy.*				
2. The system fits well with my work practices related to meat selling or selection. *Ang sistema ay angkop sa aking mga gawain sa pagbebenta o pagpili ng karne.*				

3. I believe this technology can greatly improve pork meat grading accuracy. *Naniniwala ako na malaki ang maitutulong ng teknolohiyang ito sa pagpapabuti ng katumpakan ng pag-grado ng karne ng baboy.*				
4. Using this system is an enjoyable and beneficial experience. *Ang paggamit ng sistemang ito ay isang kasiya-siya at kapaki-pakinabang na karanasan.*				
5. I would recommend this system to other pork vendors or consumers. *Irekomenda ko ang sistemang ito sa ibang mga tindero o mamimili ng karne ng baboy.*				
Behavior (Intention to Use the System)				
1. I intend to use this system regularly for pork meat grading. *Balak kong gamitin ang sistemang ito nang regular para sa pag-grado ng karne ng baboy.*				
2. I am likely to rely on this system to help with meat quality assessment in the future. *Malamang na aasa ako sa sistemang ito para makatulong sa pagtatasa ng kalidad ng karne sa hinaharap.*				
3. I am woulding to adopt other AI-based solutions for meat management based on my experience with this system. *Handa akong gumamit ng iba pang AI-based na solusyon sa pamamahala ng karne batay sa aking karanasan sa sistemang ito.*				
4.I am likely to recommend this application to colleagues or customers. *Malamang na irekomenda ko ang aplikasyon na ito sa aking mga kasamahan o mamimili.*				
5. I am committed to using this system as part of my daily meat grading routine. *Nais kong gamitin ang sistemang ito bilang bahagi ng aking araw-araw na pag-grado ng karne.*				

_____

Signature of the Respondent

APPENDIX B

Approval & Communication Letters

APPROVAL SHEET

The thesis entitled "**MEATCHECKER: AN ANDROID-BASED MOBILE APPLICATION FOR PORK MEAT GRADING UTILIZING YOLOv8 ALGORITHM"** prepared and submitted by **Michael M. Delos Santos, Kyla Marie T. Sambitan, and Jay vee O. Valera** in partial fulfillment requirements for the degree Bachelor of Science in Information Technology, specialized in Web and Mobile Application Development, is hereby recommended for approval and acceptance.

SHERWIN B. SAPIN
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of____.

MIKE PHILIP T. RAMOS
Member

CAROLINA R. JOVAL
Member

JOCELYN O. PADALLAN
Member

JEFFERSON L. LERIOS, DIT
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, specialized in Web and Mobile Application Development.

LOYD S. ECHALAR, DIT
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, EdD.
Chairperson, Research and Development Office

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

Consent Letter for Gathering Data

Laguna State Polytechnic University
56PJ+PP9,
Los Baños, Laguna, 4030
08-30-25

To Whom It May Concern,

Good day! We, the researchers of 4th Year students of Bachelor of Science in Information Technology, are currently conducting a study entitled “MeatChecker: An Android-Based Mobile Application for Pork Meat Grading Utilizing YOLOv8 Algorithm.” As part of our data collection, we respectfully request your permission to take photos of raw pork meat cuts available in your stall for research purposes.

The gathered images would be used solely for academic and research-related analysis, specifically to train and evaluate our pork meat grading system. Rest assured that no personal information about the vendors would be collected, and all data would be handled with strict confidentiality.

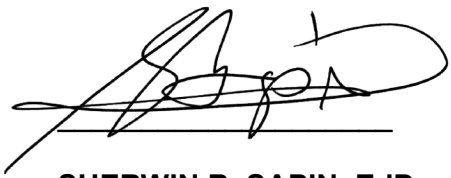
Your participation is completely voluntary, and you may withdraw at any time without any consequences.

Should you have concerns or need clarification, feel free to contact us on 09925329007 or our adviser Mr. Sherwin B. Sapin.

Thank you very much for your time and kind cooperation.

Respectfully yours,
Michael M. Delos Santos
Kyla Marie T. Sambitan
Jay vee O. Valera
Researchers / IT Students
09925329007

Noted and approved by:



SHERWIN B. SAPIN, EdD
Research Adviser

Consent Letter for Survey

Laguna State Polytechnic University
56PJ+PP9,
Los Baños, Laguna, 4030
11-12-25

Antonio Tolentino
Supervisor III
Mercado De Calamba
Calamba City

Dear Mr. Antonio Tolentino,

Good day! We are IT students from Laguna State Polytechnic University currently conducting a study titled “MeatChecker: An Android-Based Mobile Application for Pork Meat Grading Utilizing YOLOv8 Algorithm.” As part of our data-gathering process, we respectfully request permission to conduct a survey with your consumers/vendors to collect insights that would support the development and accuracy of our system.

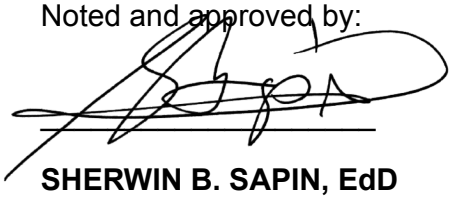
All information gathered would be used strictly for academic purposes and would remain confidential. The survey would only take a few minutes and would be conducted at your most convenient time.

This research is under the supervision of our adviser, Mr. Sherwin B. Sapin.

We sincerely hope for your kind approval and cooperation. Thank you very much, and we look forward to your positive response.

Respectfully yours,
Michael M. Delos Santos
Kyla Marie T. Sambitan
Jay vee O. Valera
Researchers / IT Students
09925329007

Noted and approved by:



SHERWIN B. SAPIN, EdD
Research Adviser

APPENDIX C

7 Pager IMRAD Format

MEATCHECKER: AN ANDROID-BASED MOBILE APPLICATION FOR PORK MEAT GRADING UTILIZING YOLOv8 ALGORITHM

Michael M. Delos Santos, Kyla Marie T. Sambitan, Jay Vee O. Valera
College of Computer Studies (CCS), Laguna State Polytechnic University, Los Baños, Laguna, Philippines

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ABSTRACT

MEATCHECKER: AN ANDROID-BASED MOBILE APPLICATION FOR PORK MEAT GRADING UTILIZING YOLOv8 ALGORITHM was developed, which would help consumers and sellers to recognize pork meat cuts and understand how fresh the cuts are. Poor inspection of pork meat quality, if not done properly, may lead to food safety risks. The goal of this app is to offer a quality assessment system that is reliable and easy to apply for pork meat images and classify them as Fresh or spoiled with machine learning analysis. Background: This article aimed to build a mobile system that can identify pork meat cuts and evaluate the freshness of cuts, implementing the YOLOv8 model on the app for use in an Android-based system for real-time classification. The application was implemented in Roboflow, Google Colab, Android Development Studio, and Kotlin, as well as TensorFlow. The system reads taken photographs, classifies pork meat cuts, and creates bounding boxes to assign freshness scores and confidence ratings. The Technology Acceptance Model (TAM) would be applied to measure perceived usefulness and ease of use by participants in order to measure user perception. Researchers found the usefulness of the application to be strongly satisfactory, based on ISO 25010 software quality standards. This gives Meatchecker: A Mobile Application For Pork Meat Grading Based on Android, an Application using the Yolov8 Algorithm, the ability to assess whether pork meat is fresh or spoiled easily for the user.

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I. INTRODUCTION

Pork is not just the most consumed meat in the Philippines, but also the fundamental ingredient of Filipino cuisine and culture. Kasim (shoulder), Liempo (belly), Pork chop (loin), and Tenderloin (lomo) are the most abundant cuts of pork utilized in domestic meals and indigenous foods nationwide. Tacitly tender, versatile, and flavorful, these cuts are popular ingredients that anyone who doesn't get this lucky enough is unlikely to use them on a plate. Soaring popularity and use make them suitable for

regular meat grading practices. Pork had been chosen over beef, chicken, and other poultry as they are the staple food in Filipino diets and are economically crucial. Also, pork is more susceptible to spoilage than other meats when refrigeration is tight in traditional wet markets. This not only makes grading freshness practical but also necessary. By looking at the four types of pork cuts, this study takes into consideration actual consumption patterns and provides real-life, relevant applications.

According to the statistics of Statista (2025), pork is still the most-consumed meat per capita in the country in 2024, exceeding both poultry and beef. Moreover, Rotecna (2023) explains that the pork industry in the Philippines has taken great strides to recover from the worst effects of the African Swine Fever (ASF) outbreak, supporting the sector's significance in the Philippine food supply and economy. The millions of backyard hog raisers and modest sellers who work in the pork sector account for at least a third of the national food surplus. Further, Veterinaria Digital S.A. (2022) reiterated that the swine trade is a major source of economic activity, noting that the product is still also a "big business in terms of jobs and consumption," even after the losses caused by ASF and other problems in the past. Advanced technologies, like computer vision, deep learning (e.g., YOLO), and spectroscopy, are explored for meat quality assessment around the world. Automated meat-grading systems, which have recently been implemented in developed nations, also provide significant efficiency since much of the quality testing relies on manual testing. Such old practices still prevail in the Philippines, thus the requirement for an easy and tech-enabled grading system to make pork quality and safety a good priority for customers. Country – Pork meat grading in the country is also affected by outdated, unreliable, subjective, and manual inspection of pork meat. Without access to updated grading technologies, small vendors can find it impossible to adhere to these standards. The process is also slow and laborious, which impacts the supply chain. Computer vision and deep learning models like YOLOv8 hold great promise; however, they are unable to be put to use or deployed online in real time due to their computational limitations.

Moreover, uncertain freshness measurement creates potential food spoilage and fraud risks. Currently, there is a specific technology gap for the accurate and effective grading of pork. The bad state of the industry comes from the examination process, as it must be carried out by hand, which causes them to lack regularity, their own personal bias, and errors during grading are more common. Additionally, small-scale vendors and butchers have limited access to new grading tools, so maintaining consistent quality and industry standards has been a challenge for these groups.

Computer vision systems and object detection algorithms (i.e., YOLOv8) have been used in food quality assessment, but mobile-based or immediate delivery of pork meat grading solutions have not come together. As an issue, current systems are either too expensive, require more complex equipment than a large number can support, or use them as only used on large-scale process plants. For this gap in technology, food safety, consumer trust, and fair trade practices are all seriously undermined; this creates serious problems for food producers that people simply cannot change or even eat. With no easy-to-use automatic system to grade meat, vendors fail to grade their products efficiently, and consumers are left with products that are either misclassified at the retail or of lower quality than normal.

Vendors and butcher firms, in general, might not have an advanced grading system, so they have a difficult time keeping quality standards and controls. Increasing consumer demand for safe, high-quality meat-especially among the pork industry, means a better grading system is needed soon that would produce safer quality meat. Computer vision and deep learning methods such as YOLOv8 have shown great potential for automating food quality control. And yet, most current grading systems are set up for large-scale meat processing operations, without low-cost and low-access solutions. This study is important to examine variations and inefficiencies in conventional pork meat grading, based on human-interpreted examination. Such commercially available systems tend to be expensive and out of reach for small vendors, which leads to a technological gap in the industry.

This project introduces an Android Grading system for Android as a low-cost, high-accuracy, and affordable option by implementing a mobile-based grading system targeted at mobile, which is our target user platform. Through computer vision and YOLOv8-Object Detection technology, this method also makes it possible to have an objective and accurate assessment of meat quality, therefore promoting the enhancement of food safety, consumer confidence, and the effectiveness of the process. This innovation would allow small vendors to create a seamless link with up-to-date grade technology, making

quality control less expensive and more standardized. In the local context, particularly in wet markets and small retailers, pork meat is often exposed for sale without environmental controls, so visual inspection (as opposed to a standardized environmental control) has become the primary method of assessing freshness and quality. People often use color, surface moisture and looks, while vendors use experience, not calibrated tools. These customs are likely to make the grading inconsistent and contribute to the accidental spread of spoiled meat. The development of a mobile-based grading system in the study locates technology in daily market functions that place a premium on affordability, portability, and user-friendliness. The proposed solution meets national food safety objectives as it enables stronger decision-making right at the point of sale. It also promotes the use of technology by small-scale food vendors, transparency in selling meat, and protection of public health. The incorporation of an Android-based pork meat grading application is therefore more than just technical, and rather closes the economic, and social gap, cementing sustainable food initiatives as well as creating confidence in the pork supply chain. This background also signals the need to apply AI at the community level, where most pork transactions are transacted on a daily basis. Instead of trying to supplant human judgment, the system becomes a decision-support tool and helps the vendor get smarter with the right answers based on data. It can help minimize buyer/seller disputes, facilitate inspections of marketplaces, and offer educational value through raising the awareness of indicators of meat quality. Finally, the mobile technology enables scalability and future integration with cloud databases, reporting and policy monitoring structures. By integrating cutting-edge computer vision methods with real markets, the research showcases how innovation can be directly employed for the benefit of consumers, end users, and/or the overall food supply chain. This additional context enhances the study's applicability to national development, food securit,y and inclusive technological advance. It focuses on feasibility, real-world impact, and adoption in the long run.

II. RELATED WORKS

Pork grading systems differ from country to country and cater to enhancing product performance as well as satisfying consumer demands. An updated grading system in India was established, whereby pig types received yield and quality grades in relation to the current grading system, which aimed to minimize the deficiencies of the original one (Thomas et al., 2018). Previous research on the quality of Korean pork carcass has reported higher grading to relate to the greater fat content, unsaturated fatty acids, and flavor compounds, resulting in high sensory quality (flavor, juiciness, and tenderness) (Van-Ba Hoa et al., 2019). These results emphasize how impactful grading systems are to market needs and consumer satisfaction.

Computer vision systems (CVS) have become an effective substitute for standard methods in evaluating meat quality, providing gentle, fast, and economical solutions (Taheri-Garavand et al., 2019; Modzelewska-Kapituła & Jun, 2022). These technologies can assess multiple quality attributes, such as color, acidity (pH), moisture levels, and microbial presence, through digital imaging techniques (Rahman et al., 2020). CVS applications include categorizing fresh and processed meats, examining carcass composition, and evaluating the effects of technological processes on meat characteristics (Modzelewska-Kapituła & Jun, 2022). Different methods have been explored for feature removal and classification, including GLCM and KNN (Hanggara et al., 2021), intense learning and OpenCV (Lee et al., 2022), and thresholding with decision tree organization (Adi et al., 2017). Key quality clues assessed include marbling, meat color, and fat color (Adi et al., 2015). Similar techniques have been applied to separate between beef and pork, with one study reporting over 89% correctness using textural analysis and neural networks (Muhammadiy & Fahmi, 2016). The applications are designed for smartphones, allowing real-time meat quality examination. These systems aim to provide objective and efficient alternatives to traditional visual inspection methods, addressing concerns about meat quality and possible fraud in the market (Muhammadiy & Fahmi, 2016; Adi et al., 2015).

Current improvements include the application of object detection algorithms like YOLOv8 NAS for multi-cuisine quality filtering at service points (Kosuru et al., 2023). These systems might be used in place of manual inspection, increasing efficiency and reliability in food quality evaluation (Ambili M & Chandren, 2021). As these technologies continue to evolve, they are predicted to be used widely in an automated production process to provide quality assurance in a competitive food industry and are also likely to be extended to consumer-grade quality review on mobile devices in the coming years (Külcü, 2018). Advanced learning in particular is potentially viable for on-the-fly application in the food

industry, given its feature learning capabilities (Lin et al., 2022). Harmless technologies such as acoustic analysis, machine vision, electronic nose, and spectral imaging, in concert with high-tech machine learning methods, have brought promising results in food quality evaluation (Lin et al., 2022).

Recent research has focused on enhancing real-time detection of pork meat using YOLO-based models. Several studies have proposed advancements to YOLO architectures for pig detection, including the TR-YOLO model, which includes self-attention mechanisms and large layered kernels (Shihua Pu & Zuohua Liu, 2024), and the YOLOv3-SC model, which merges attention modules and distributed pyramid pooling (Wangli Hao et al., 2022). This progress has led to increased detection, accuracy, and effectiveness. Another study introduced an improved YOLOv3 model utilizing transfer learning and attention methods, achieving an average precision of 94.12% (Dan Yin et al., 2021). In addition, a novel approach based on YOLACT was developed for the real-time detection and tracking of pig body parts, achieving 90% validity and a processing speed of 30.5 fps (Fuen Chen et al., 2020). The following advancements provide more effective monitoring of pig health, behavior, and welfare in intelligent breeding systems.

IV. RESULTS AND DISCUSSION

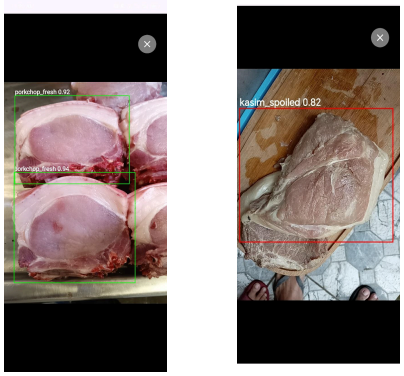


Figure 3. Detection Result

As shown in Figure 16, the system revealed the mobile application's capability to recognize and classify pork meat freshness with the application of color-coded indicators, with green indicating freshness and red indicating spoiled status. The application screened, uploaded, or captured images, determined the cut type of meat, and then displayed a confidence score. The kasim was declared to be fresh while the liempo was tagged as spoiled. This process aligned with the study's goal of helping consumers assess pork freshness and preventing the purchase of spoiled meat through real-time image analysis.

As shown in Figure 8, the code imported a YOLOv8 model converted to TensorFlow Lite, activated it with the TFLite interpreter, and verified the expected shape and type of data entered by the model to determine whether it was suitable for inference. It scanned the whole image for regions with consistent shapes with known objects. If a potential object was identified, a bounding box would be drawn around that portion as well.

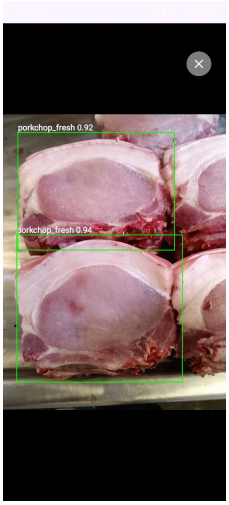


Figure 9. Multiple Object Detection with Classification

As shown in Figure 9, the whole image was scanned by the model to generate areas corresponding to the shape of the known objects. When a potential object was discovered, a bounding box was drawn around that region. The model then considered each boxed region and classed them, for example, "kasim_fresh," "porkchop_fresh." Finally, a confidence score was defined, indicating the model's certainty of prediction, and numerous objects were detected and labeled in a single image.

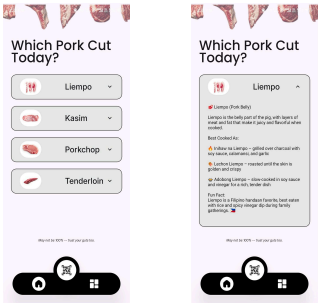


Figure 10. Home Screen

As shown in Figure 10, the Home Screen of the MeatChecker application was displayed. The screen also provided dropdown sections for pork cuts, beside the "Scan Now" button. Once the user extended a dropdown, it showed a quick snippet of the exact pork cut, along with the part of the pig, the optimal cooking method, and a fun fact. This enabled the user to learn more about the characteristics of pork meat before any scan. When ready, the user clicked on "Scan Now," which pushed him to start up his camera for real-time scanning using the YOLOv8 model. The layout guaranteed an informative and user-friendly interface that would be appropriate for both consumers and vendors.

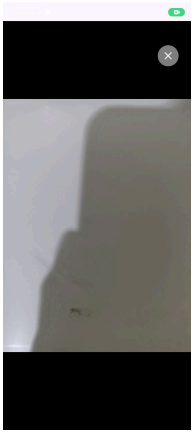


Figure 11. Camera Scanning Screen

As shown in Figure 11, the Camera Scanning Screen appeared. When the device tapped "Scan Now," it opened the camera window and showed an actual preview of what it'd seen live. When the user snapped any image of pork, the image was sent to the YOLOv8 model, which analyzed and ranked the meat's freshness. This was also the screen where the detection process officially began.

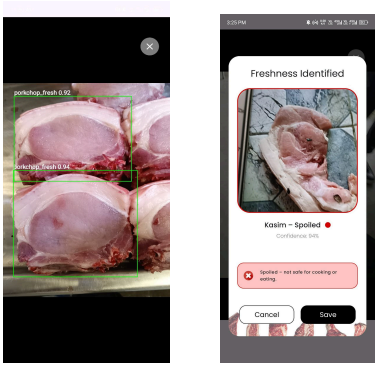


Figure 12. Detection Result and Preview Screen

As shown in Figure 12, the detection results after scanning were displayed by the system. It identified the pork cut, presented freshness classification, and the confidence percentage yielded by the YOLOv8 model. Higher confidence scores (80–100%) implied more certainty in the model, and lower values indicated more doubt as a consequence of variations in lighting, angle, or varying degrees of occlusion in the meat. Before saving, the user reviewed the image and read some recommendations for cooking the item. The report could then be canceled or saved by the user.

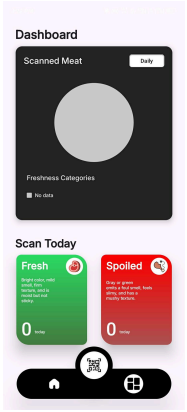


Figure 13. Dashboard (Reports Screen)

As shown in Figure 13, the Dashboard screen. This page serves as the report section of the application, where users can view a summary of their scanned pork results. The dashboard displays daily scanned data, showing the number of pork samples classified as Fresh and Spoiled based on the YOLOv8 detection results. Each category includes a brief description of the meat characteristics and the number of times that classification occurred on that day. This allows users to monitor and review their scanning activity easily.

Test and Evaluate the Software using ISO 25010

ITEM	MEAN	STANDARD DEVIATION	INTERPRETATION
Functional Completeness	4	0	Strongly Agree
Functional Correctness	3.33	±0.58	Strongly Agree
Functional Appropriateness	3.33	±0.58	Strongly Agree
Time Behaviour	4	0	Strongly Agree
Resource Utilization	4	0	Strongly Agree
Capacity	3.67	±0.58	Strongly Agree
Co-existence	3.67	±0.58	Strongly Agree
Interoperability	3.67	±0.58	Strongly Agree
Appropriateness	3.67	±0.58	Strongly Agree

Recognizability			
Learnability	4	0	Strongly Agree
Operability	4	0	Strongly Agree
User Engagement	3.67	±0.58	Strongly Agree
Inclusivity / Accessibility	4	0	Strongly Agree
Availability	4	0	Strongly Agree
Fault Tolerance	3	0	Agree
Recoverability	3.67	±0.58	Strongly Agree
Consistency	3	0	Agree
Confidentiality	4	0	Strongly Agree
Authenticity	4	0	Strongly Agree
Resistance	4	±0.58	Strongly Agree
Analysability	3.33	±0.58	Strongly Agree
Modifiability	3.33	±0.58	Strongly Agree
Adaptability	3.67	±0.58	Strongly Agree
Scalability	3.33	±1.15	Strongly Agree
Installability	4	0	Strongly Agree
Replaceability	3.67	±0.58	Strongly Agree
Overall	3.69		Strongly Agree

Table 1. Results of Beta Testing answered by the IT Professional using ISO 25010

As shown in Table 1, IT experts reported the minimum beta test result to be 3.00, which indicates a verbal response of "Agree." By contrast, the highest mean was 4.00 (Strongly Agree). The value of the overall mean, 3.69, suggested a general agreement among the IT experts on the system. These combined observations showed that the system was satisfactory, meeting the expectations and standards in beta testing, and met the approval requirements.

Test and Evaluation Performed using TAM

The results are collected from the respondents, who are the consumers and vendors that would use the application.

ITEM	MEAN	STANDARD DEVIATION	INTERPRETATION
The system improves the efficiency of grading pork meat based on freshness and quality.	3.60	±0.85	Strongly Agree
The system provides accurate and real-time grading results that support better decision-making in meat selection.	3.08	±0.80	Agree
Using this system increases productivity and ensures consistency in pork meat quality assessment.	3.19	±0.77	Agree
The system helps identify meat quality issues before the product is sold or consumed.	3.19	±0.77	Agree
I find the system valuable in ensuring high-quality pork meat for vendors and consumers.	3.17	±0.73	Agree
Overall	3.17		

Table 2. Level of Perceived Usefulness

As shown in Table 2, the table reflects how users would rate the pork meat grading system. A mean of 3.60 reflected "Strongly Agree," which indicated that users agreed the system greatly enhanced efficiency in pork meat grading by freshness and quality. Users also reported that as a result, the system produced accurate, real-time grading results that guided better judgment in meat selection, with an average score of 3.06 and a mean score of "Agree." Moreover, pork quality assessment was also seen by the system as increasing productivity and maintaining consistency in pork meat grading, with a mean of 3.19. The same results showed that the system contributed to detecting meat quality concerns before the product was sold or consumed, with a mean score of 3.19. Users also stated the system was valuable to them so they can ensure there was high-quality pork meat for both vendors and consumers, on average scoring 3.17. In total, respondents mainly "Agreed" to "Strongly Agreed" towards the functional, reliable, and useful aspect of the system for achieving correct and regular grading of pork meat.

Result of Software Testing

Unit Testing Result

Unit testing for the Home Screen or Dashboard module was successfully completed. The interface displayed correctly, including the "Scan Now" button and pork-cut dropdowns, and navigation to the Camera Scanning Screen worked smoothly.

Unit testing for the Camera Scanning Screen module also passed. The live camera preview appeared correctly after tapping "Scan Now," and the real-time scanning feature worked smoothly using the YOLOv8 model, producing accurate analysis results.

Unit testing for the Detection Result and Preview Screen module was likewise successful. The system displayed accurate detection details such as pork cut, freshness classification, confidence level, and bounding box markers and the preview, save, and cancel functions operated correctly.

Unit testing for the Dashboard or Reports Screen module was also successfully passed. The summary of scanned classifications displayed accurate data, the visualization of daily and weekly counts worked properly, and navigation back to the Home Screen functioned without issues. Reporting and tracking system works properly.

Integration Testing Result

Integration testing requires the integration of all modules that contribute to the system's functionality. Connecting the Home Screen module, Camera Scanning Screen module, Detection Result and Preview Screen

module, and Reports / Dashboard Screen module enhances the overall performance of the MeatChecker application. This testing ensures that all modules communicate effectively, data is transferred accurately, and the user experiences a smooth and continuous workflow from scanning to viewing results.

System Testing Result

System testing of the application. The testing was conducted manually using a physical Android device to verify the application's overall functionality and performance in an offline environment. Since the MeatChecker application functions without internet connectivity, all modules were tested to ensure proper operation of scanning, detection, and report generation features. The results confirmed that the system performs efficiently and consistently, with response times ranging from 0.5 to 3 seconds, ensuring its readiness for deployment. This finding revealed that the system testing received a positive result, with all modules performing as expected.

Acceptance Testing Result

Acceptance testing characteristics are functionality, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability, with interpretations of 'Strongly Agree'. The acceptance testing results indicate that the software meets the acceptance criteria of the chosen respondents, based on the users' roles and how they use the application. Acceptance testing enables developers to engage end users in the testing process and gather their feedback through ratings, helping developers understand the needs of each function in the tested software.

V. CONCLUSION

In conclusion, this application can classify pork meat by analyzing its image and identifying whether it is Fresh and Spoiled. It successfully detects different pork cuts and provides a corresponding freshness classification that can assist both consumers and vendors in evaluating meat quality. The application allows users to gain information about the condition of pork meat through automated detection, offering a practical and convenient tool for real-time meat assessment. Based on the ISO 25010 software quality standard, the system complies with user needs in terms of functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. The researchers concluded that the developed Android application was rated strongly acceptable under the ISO 25010 characteristics of software quality. This implies that the use of MeatChecker makes it easier for users to assess pork meat quality and supports safer and more informed decision-making when handling or purchasing pork meat.

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APPENDIX D

Plagiarism Result

Report: CHAP 1 to 2 MEATCHECKER

CHAP 1 to 2 MEATCHECKER

by Ronah joy Ramirez

General metrics

66,597	9,923	588	39 min 41 sec	1 hr 16 min
characters	words	sentences	reading time	speaking time

Score

99

This text scores better than 99% of all texts checked by Grammarly

Writing Issues

22	2	20
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

CHAPTER 4 TO 5 MEATCHECKER

by Ronah joy Ramirez

General metrics

34,847	5,150	371	20 min 36 sec	39 min 36 sec
characters	words	sentences	reading time	speaking time

Score

99

This text scores better than 99% of all texts checked by Grammarly

Writing Issues

17	1	16
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

MeatChecker_ An Android-Based Mobile Application for Pork Meat Grading Utilizing YOLOv8 Algorithm.docx

ORIGINALITY REPORT

8%	5%	3%	5%
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11	Submitted to Our Lady of Fatima University-Valenzuela	<1%

APPENDIX E

Curriculum Vitae



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TECHNICAL SKILLS

- Can code in HTML/CSS, Javascript and PHP.
- Knowledgeable in using MySql.
- Knowledgeable in Photoshop and Figma.
- Can design user intergace (UI) front end.

LANGUAGES

- English
- Filipino

MICHAEL M. DELOS SANTOS

Motivated entry-level web developer with a strong willingness to learn, seeking an opportunity to grow and contribute in a collaborative development environment.

EDUCATIONAL BACKGROUND

Tertiary

Bachelor of Science in Information Technology
Laguna State Polytechnic University (LSPU) — 2022-2026
Barangay Malinta, Los Baños, Laguna

Secondary

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Primary

Mayapa Elementary School — 2010-2016
Brngy. Mayapa, Calamba City, Laguna



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Calamba City, Laguna

PERSONAL SKILLS

- Communication
- Adaptability
- Time Management
- Collaboration

TECHNICAL SKILLS

- C#
- HTML/CSS

LANGUAGES

- English
- Filipino

KYLA MARIE T. SAMBITAN

I'm eager to learn, grow, and apply my skills in a place where I can contribute and improve. I'm motivated to work in an environment where I can challenge myself and be part of something meaningful.

EDUCATION BACKGROUND

Primary

Florentino Joya Elementary School (2008 -2014)

junior

Palo Alto Integrated School (2014 -2019)

Senior

Palo Alto Integrated School (2019 -2021)

Tertiary

Laguna State Polytechnic University (PRESENT)



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TECHNICAL SKILLS

- JAVA, HTML, CCS
- C#
- Microsoft Office Word
- Microsoft Office PowerPoint
- Microsoft Office Excel

LANGUAGES

- English
- Filipino

JAY VEE O. VALERA

Secure a position in a development and dynamic work atmosphere where I can place my programming and web skills to use

EDUCATIONAL BACKGROUND

Tertiary

Bachelor of Science in Information Technology
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Calamba Bayside Integrated High School — 2020-2022

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Calamba,

Junior High School

Calamba Bayside Integrated High School — 2016-2020

Mt. Halcon St. Barangay Palingon,
Calamba,

Primary

Jose Rizal Memorial Elementary School — 2010-2016

I hereby attest that the information provided above is true and correct to the best of my knowledge and ability.

JAY VEE O. VALERA